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Quantifying Regional Variation in Ivermectin Use During the Pandemic using Regularized Synthetic Controls

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Abstract

We analyze weekly U.S. prescription claims data over a 36-month period spanning the COVID-19 pandemic to investigate patterns of overconsumption of the antiparasitic drug Ivermectin (IVM). To quantify the overconsumption of IVM use following the heightened public attention on IVM as a candidate treatment for COVID-19, we adopt a causal framework based on synthetic controls, comparing IVM prescription trends to those of a large set of control medications. We employ a regularized synthetic control (RSC) method using continuous spike-and-slab shrinkage priors to estimate state-level deviations in IVM consumption. This approach offers decision-theoretic guarantees on predictive risk, supporting its application in downstream policy analysis at multiple-time points after the intervention. Its empirical robustness is demonstrated through extensive validation checks. Our findings reveal a modest increase in IVM prescriptions following early reports of its potential therapeutic use, with no significant surge over the subsequent eight months. This was followed by a pronounced increase that coincided with the peak in COVID-19 cases. Strikingly, elevated IVM use persisted even after COVID-19 vaccines became widely available and multiple federal countermeasures were implemented. Our estimation approach captures the heterogeneity in long-term effectiveness of these countermeasures across states. We find that state-level political affiliation significantly explains variation in overconsumption, even after accounting for COVID-19 incidence rates. These findings highlight deep regional disparities in the effectiveness of public health messaging and suggest a need for more targeted and trusted communication strategies.

Key words: drug misuse, heath-care communications, Ivermectin, overconsumption, persistence, regularized synthetic control, spike-and-slab priors.

1. Introduction

During the COVID-19 pandemic, several pharmaceutical treatments attracted widespread public attention and experienced substantial increases in consumption—often exceeding clinically recommended levels (Perlis et al., 2023; Vaduganathan et al., 2020). One such drug was Ivermectin (IVM), an antiparasitic medication primarily used to treat parasitic infections (Hill et al., 2022). Overuse of IVM can lead to adverse health

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outcomes, prompting multiple federal regulatory agencies to issue warnings against its misuse (Bray et al., 2020). Nevertheless, IVM consumption remained elevated throughout the pandemic.

While some exploratory or prophylactic use is expected during a frightening public health crisis, the persistence and magnitude of IVM overuse raise critical questions about the effectiveness of medical countermeasures and health communication strategies. Accurately quantifying IVM overconsumption requires a rigorous causal inference framework (Imbens, 2024) that accounts for confounding factors such as seasonality, temporal trends, and concurrent COVID-19-related interventions (e.g., shelter-in-place orders, mask mandates, and travel restrictions), which may have influenced drug purchasing behavior.

In this study, we analyze weekly prescription claims data in the United States over a 36-month period corresponding to the COVID-19 pandemic to examine patterns of IVM overuse. Prescription claims data serve as a widely accepted proxy for drug consumption and have been employed in prior studies to investigate medication overuse (Vaduganathan et al., 2020). To establish a baseline, we compare IVM prescription trends to those of a control group of medications that did not receive public or media attention as potential COVID-19 treatments. We apply synthetic controls (SC) (Abadie and Gardeazabal, 2003; Abadie et al., 2010) to estimate counterfactual prescription levels for IVM, isolating the overconsumption attributable to pandemic-era attention. The resulting average treatment effect on the treated (ATT) captures the incremental change in IVM use due to heightened attention.

We construct state-level ATT estimates to map IVM overuse across the U.S. over time, revealing substantial regional and temporal heterogeneity. These findings point to structural differences in institutional trust and the effectiveness of public health messaging, underscoring the need for more targeted and resilient health communication systems in future public health crises.

1.1. IVM Over-consumption: Background and Timeline

Even though Ivermectin (IVM) first received scientific attention in the context of COVID-19 from an Australian study in April 2020, which was subsequently retracted, it was not widely known outside the scientific community at that time. IVM gained significant attention as a potential treatment for COVID-19 after Dr. Pierre Kory's testimony at a US Senate hearing on December 8, 2020, where he advocated for its use. Subsequently, IVM received considerable attention in conservative media and from celebrities. The initial evidence supporting IVM was eventually discredited. The countermeasures to inform public about IVM misuse were primarily informative (Borges Nascimento et al., 2022), with no imposed restrictions on its prescription for COVID-19 treatment. Further evidence emerged indicating the ineffectiveness of IVM against COVID-19, and several studies that had brought IVM into the spotlight for COVID-19 were discredited by July 2021 (Hill et al., 2022).

Table 1. Timeline of Ivermectin (IVM) related events

Event Id	Date	Event
1	04/01/2020	Australian Study Suggests IVM Could be a Treatment for Covid-19
2	12/08/2020	Dr. Kory testifies in favor of IVM in a senate committee
3	01/14/2021	NIH: Not enough data to say IVM works
4	02/04/2021	Merck issues a warning against IVM Use
5	03/05/2021	FDA warns against IVM use
6	07/04/2021	67% of adults in the US receive at least one dose of vaccine
7	07/15/2021	Key studies that drew attention to IVM discredited
8	08/26/2021	CDC issues warning on IVM use
9	12/31/2021	End of analysis period

We present the timeline of IVM-related events in Table 1 and the plot of the weekly IVM prescription claims (indexed) in the US in Figure 1. Although the early study in April 2020 offered some preliminary evidence favoring IVM use against COVID-19, Figure 1 shows that IVM prescription claims increased substantially only in December 2020, coinciding with a U.S. Senate hearing exploring alternative treatment

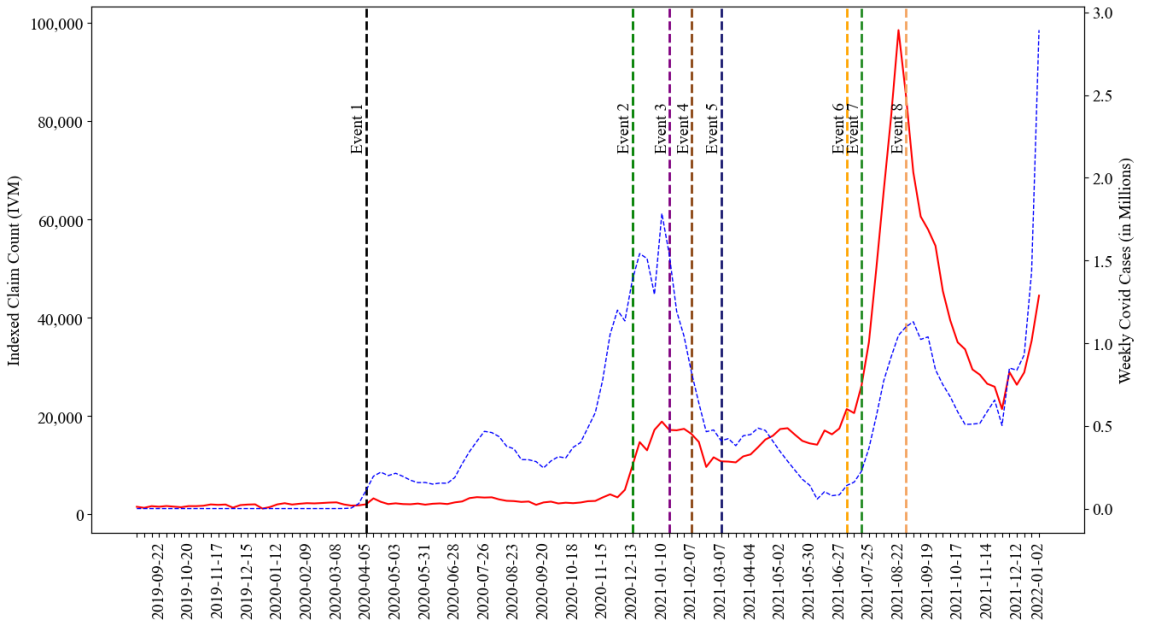

Fig. 1: Plot of total US (indexed) IVM prescription claims (in red) and Covid case counts (in dotted blue) at weekly resolution. The weeks are indexed by their start date on Sundays. The time points marked by dotted vertical lines correspond to major related events listed in Table 1.

strategies (Bonis and Curtis, 2020). The National Institute of Health (NIH) issued a formal statement on January 14, 2021, stating that there was insufficient evidence supporting IVM's effectiveness against COVID-19 (National Institute of Health, 2021). However, prescription claims began to rise again in August 2021, coinciding with an increase in COVID-19 cases. By that time, 67% of the U.S. population had received at least one dose of the COVID-19 vaccine[1], and the initial study promoting IVM for COVID-19 had been formally discredited (Hill et al., 2022). Interestingly, IVM claims peaked before COVID-19 cases in 2021, suggesting that some patients might have used IVM as a prophylactic. Additionally, there was an announcement by the Centers for Disease Control and Prevention (CDC) against IVM use on August 26, 2021 (Christensen, 2021). Despite this announcement, we observe an increase in IVM prescription claims in December 2021. We aim to determine the causal impact of these measures on observed changes in IVM prescription claims, while accounting for the effects of confounding factors.

1.2. Causal Study of IVM Overuse

We use the synthetic control (SC) method to estimate the net changes in IVM prescription claims due to increased public attention during Covid. We consider a very large control set of medications which provides improved control for unobserved confounders. But, we can not directly apply the widely used SC criterion of Abadie et al. (2010) for estimating the state wise ATTs for IVM as we have more controls than the number of pre-treatment observation time points. In such scenarios, it is recommended to use statistical shrinkage to produce robust SC estimates (Ben-Michael et al., 2021). Detailed literature review on this topic is provided in Sec 3.1. We employ a spike-and-slab shrinkage prior (Malsiner-Walli and Wagner, 2018; Ročková and George, 2018; Antonelli et al., 2019) based Bayesian SC (BSC) method to estimate the state-wise ATT for IVM. We provide optimal predictive risk properties of the ATT estimates. Controlling the predictive risk of ATT estimates is crucial for our application, as we are not only interested in using the ATT as an estimate of the

[1] `https://www.cbsnews.com/news/biden-covid-19-vaccine-goal-missed/`

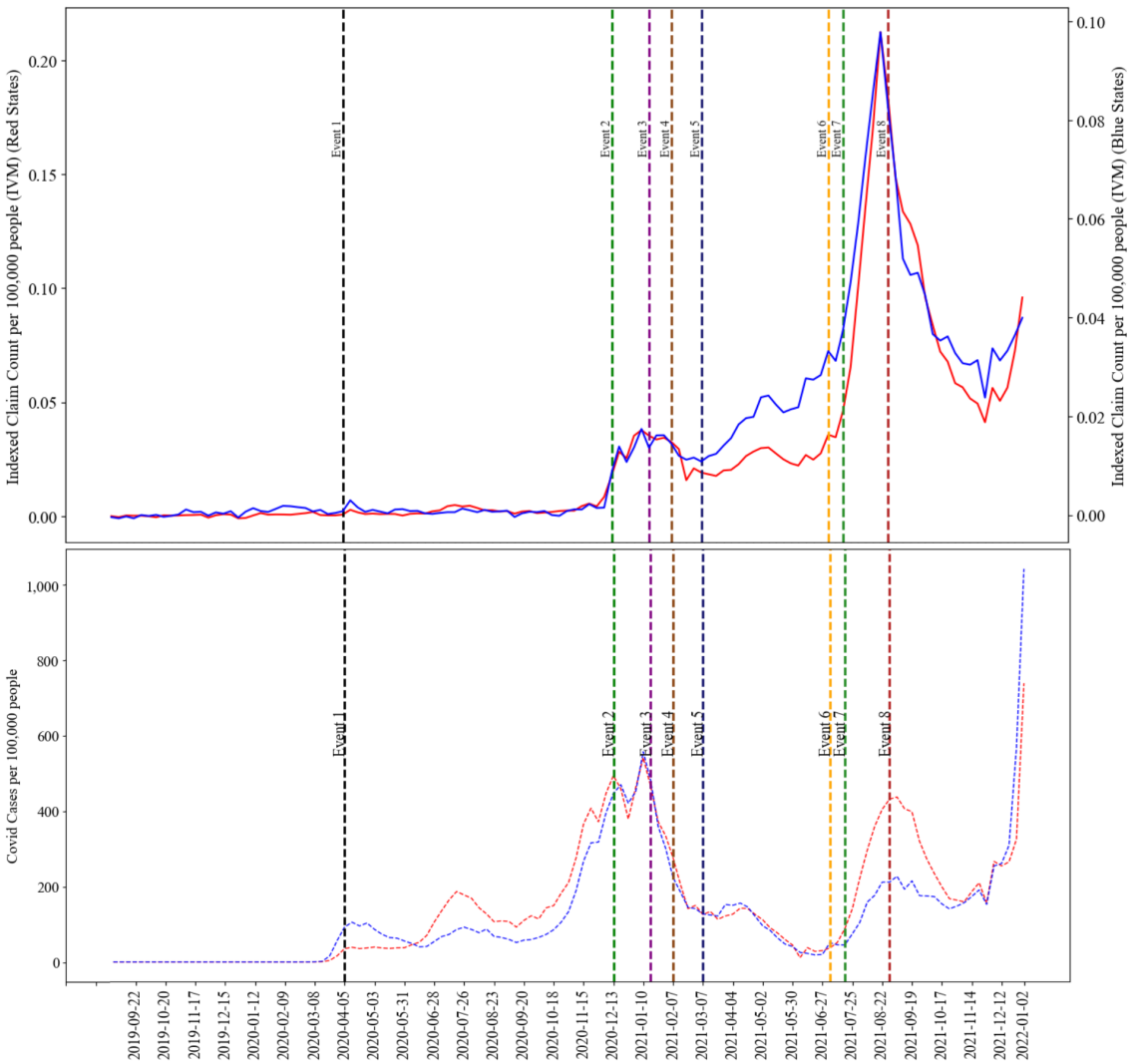

Fig. 2: Plot of weekly per capita IVM prescription claims (indexed) averaged across states that voted *Republican* (in red) and those voted *Democratic* (in blue) in the 2016 US presidential election. The time points marked by dotted vertical lines correspond to major related events listed in Table 1.

overconsumption, but also in conducting further downstream analysis to study the following two important scientific questions.

a) Measuring Persistence of IVM overconsumption. We empirically study the weekly changes in ATTs over twenty months as different countermeasures were announced to understand their impact on the persistence of IVM over-consumption. Informative countermeasures, such as regulatory advisories and debunking efforts, have shown mixed effectiveness in prior studies (Borges Nascimento et al., 2022). While some evidence suggests that the credibility of the source enhances their impact (Fong et al., 2022), other research finds that drug overuse can persist despite such interventions (Ecker et al., 2010, 2022; Lewandowsky et al., 2012).

b) Explaining State-Level Variation in Countermeasure Effectiveness. As IVM gained significant attention as a potential COVID-19 treatment—particularly in conservative media, we examine whether a state's political affiliation explains variation in its responsiveness to federal countermeasures. Prior descriptive studies have documented a strong correlation between IVM overuse and political affiliation (Barnett et al., 2022). To explore this relationship, we analyze state-level results from the 2016 U.S. presidential election, using a conservativeness index defined as the ratio of the Republican vote share to that of the nearest competitor, following standard practice (Pew Research Center, 2024).

Figure 2 plots per capita IVM consumption averaged across states that voted Republican (hereafter, Red states) and those that voted Democratic (Blue states). Red states exhibit substantially higher average consumption than Blue states (note the differing scales), although the timing of consumption peaks is broadly aligned across the two groups. However, the relationship between these peaks and COVID-19 case surges appears more heterogeneous and state-specific. Motivated by these patterns, we examine whether heterogeneity in states' responsiveness to countermeasures, measured through ATT estimates, is associated with the states' conservativeness index.

1.3. Our Contributions

We describe the main contributions of our work below:

- We employ a shrinkage prior-based Regularized Synthetic Control (RSC) method to estimate the ATTs, using continuous shrinkage priors based on the logit-normal distribution (Thomson et al., 2019). The proposed RSC approach (see Sec. 3) is computationally efficient and enables straightforward Bayesian inference for ATTs, supporting subsequent analyses of persistence and regional susceptibility.

  Within a factor model framework, we show that the ATT estimates generated by the RSC method exhibit well-controlled prediction error (see Sec. 3.1). To establish this property, we build upon theoretical results on the predictive performance of LASSO estimator (Chatterjee, 2013; Tibshirani and Wasserman, 2017) and posterior contraction results for spike-and-slab posteriors (Bhadra et al., 2016; Bhattacharya et al., 2015; Song and Liang, 2023). This predictive risk control ensures that multi-step-ahead ATT estimates can be reliably used for inference in the downstream analyses.
- We report state-level ATTs capturing the net increase in IVM consumption over a 72-week period following early reports of its potential therapeutic use. At the national level, IVM use exhibited a modest uptick following Dr. Kory's Senate testimony and remained elevated thereafter, with pronounced peaks in July 2021 coinciding with a surge in COVID-19 cases. This temporal alignment supports the interpretation in existing literature (Rockwell et al., 2025; Kautsar et al., 2025) that IVM overconsumption was partly driven by prophylactic use during periods of heightened infection risk. We empirically demonstrate that the reported ATTs are robust to a placebo test, and we rule out alternative explanations such as stockpiling, drug shortages, and differences in media coverage of countermeasures across states in the robustness section.
- We investigate the medium-term effects of IVM overuse by analyzing prescription claims up to twenty months after public interest in its efficacy began. In contrast to prior work—which has primarily focused on short-term, descriptive analyses of prescription trends (Vaduganathan et al., 2020; Barnett et al., 2022)—our study adopts a causal framework to assess sustained impacts. Remarkably, we find that IVM prescriptions remained elevated even after the initial studies promoting its use were discredited. Most notably, IVM prescribing increased by approximately 16 times after COVID-19 vaccines became widely available (see Sec. 4.2).
- We examine whether political beliefs influence the effectiveness of countermeasures. Our results indicate that regulatory messaging was less effective in politically more conservative states. We also study the incremental effect of definitive scientific evidence against the use of IVM on prescription claims (see Sec. 4.3). *Our work complements and extends recent studies of Ivermectin prescribing during the COVID-19 pandemic (Rockwell et al., 2025; Barnett et al., 2022) by using a synthetic control–based causal framework to estimate state-specific treatment effects through counterfactual prescribing trajectories over an extended observation window (see Appendix A for a detailed comparison).*

1.4. Organization of the paper

The paper is organized as follows. In Section 2, we present the details of the data used for the IVM case study. In Section 3, we describe the robust synthetic control methodology employed to estimate the state-wise ATT, reflecting the relative increase in IVM claims compared to control drugs. We also present the subsequent inferential analyses based on the functionals of the estimated ATTs and provide asymptotic guarantees on the accuracy of the ATT estimates. To understand the persistence of overconsumption and the impact of countermeasures, we conduct bulk-level as well as subgroup-level analyses based on state characteristics. In Section 4, we present our empirical results using the data set described in Section 2. We conclude with a discussion in Section 5. All proofs and detailed supporting tables, plots, and results are provided in the appendix.

2. Data

We use three different data sources: data on prescription claims, data on COVID-19 cases, and vote shares for the presidential candidates in the 2016 U.S. presidential election.

Prescription Claims Data. Our primary data for estimating the ATT for increased relative claims of IVM is obtained from GoodRx, a company that offers a telemedicine platform, as well as a website and mobile app that provide free drug coupons for discounted medications in the United States (US). We obtained state-level weekly prescription claims at the drug name level for the years 2019 (start) to 2021 (end) from GoodRx, covering all 50 states. Prescription claims fulfilled in a hospital setting are not included in our data. In other words, all our prescriptions require healthcare experts to agree to the course of treatment outside of a hospital setting, which is outside the emergency use authorization (EUA) orders issued by the FDA during the pandemic.

In Figure 3, we plot the average IVM prescription claims per capita for each states for four different time interval. We see not only high temporal variability but also significant differences across the US states. The IVM claims increased after Event 2 in Table 1. This increased claims continued even after event 6.

We have weekly claims data recorded at the drug name level, from which we identify IVM prescription claims and tabulate claims for other drugs. We broadly divide the drugs into (a) COVID-19 drugs based on NIH's treatment advisories between April 21, 2020, and December 30, 2021, and (b) non-COVID-19 drugs. For the group of non-covid drugs, we aggregate claims by GPI2 code for each week. Thus, our control group of medications consists of COVID-19 drugs and GPI2 code categories. We exclude Hydroxychloroquine drugs from our control group as they were also subject to widespread media and public interest during the pandemic (Perlis et al., 2023). We also excluded drugs and GPI2 categories with low volumes in the pre-treatment window and finally considered 81 COVID-19 drugs and 92 GPI2 categories. For presentation ease, hereon we term all 173 of them as control group drugs, keeping their aggregation based on category codes implicit.

As our data spans January 2019 to December 2021, it allows us to observe prescription claim behaviors well after scientific evidence showed that IVM is not a treatment for COVID-19 and that viable vaccine alternatives exist. To address data privacy concerns, the claims are indexed before conducting the analysis. The indexing is constructed by multiplying an arbitrary scaling factor for all values before any modeling. This masks the actual magnitude but preserves relative magnitudes, which are sufficient statistics for all our analyses. We report indexed claims in all the plots presented in the paper.

Secondary Data: COVID-19 Case Counts and Political Affiliations. To measure political affiliation of each state, we use state-level results from the 2016 U.S. presidential election from New York Times (2016). To control for the potential impact of countermeasures or policies on IVM prescriptions, we use the event timelines reported in Table 1. For COVID-19 case counts, we rely on the weekly time series data published by New York Times (2023).

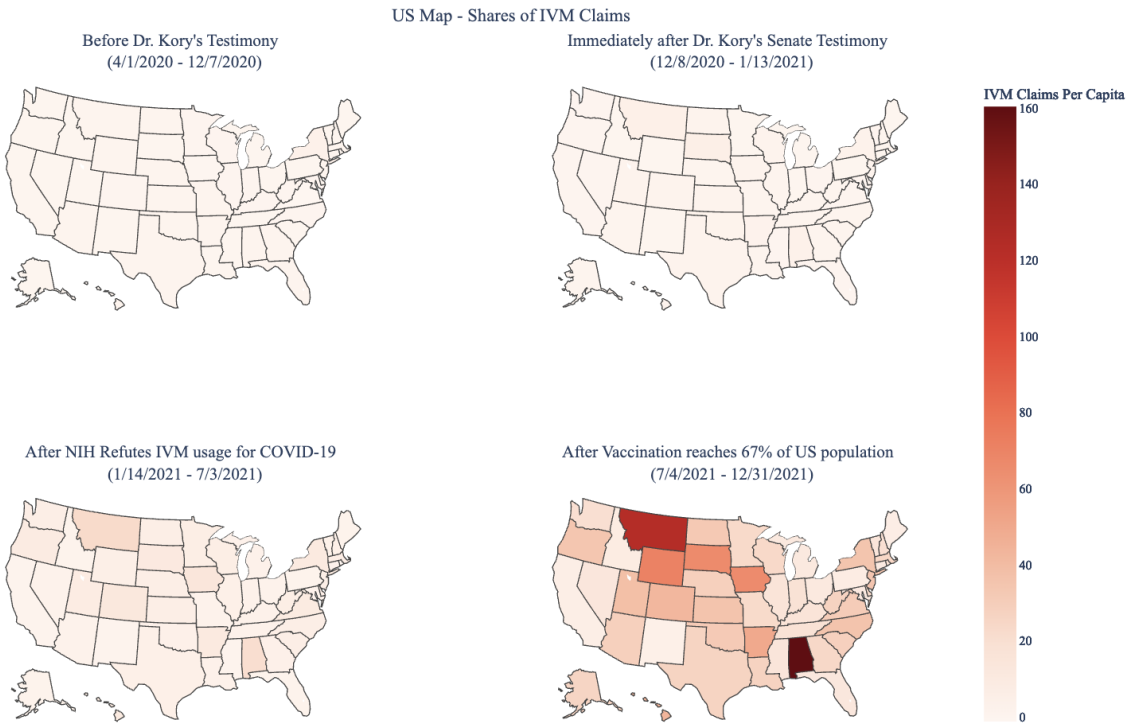

Fig. 3: Maps of average weekly IVM prescription claims for US states across four time intervals based on Table 1. From top-left, row wise we have maps for (a) Event 1 to Event 2 (b) Event 2 to Event 3 (c) Event 3 to Event 6 (d) Event 6 to Event 9.

3. Methodology

3.1. Estimating State-level ATTs using SC

Let X_{cst} denote the prescription claims of control drug $c = 1, \ldots, C$ at state $s = 1, \ldots, S$ and time $t = 1, \ldots, N$ weeks. The treatment date is April 1, 2020, the date when the first study on IVM was published online. The treatment occurs between n and $n+1$ weeks, so we have n pre-treatment weeks and $(N-n)$ post-treatment weeks. Let Y_{st} denote the observed IVM prescription claims at state $s = 1, \ldots, S$ and time $t = 1, \ldots, n$ weeks. After the intervention, we denote the observed IVM prescription claims by $Y_{st}^{(1)}$. For all $t > n$ and $s = 1, \ldots, S$, by $\widehat{Y}_{st}^{(0)}$ we estimate the counterfactual $Y_{st}^{(0)}$– IVM prescription claims had there been no intervention. Then, the ATT estimate for state s and time t is: $\widehat{\text{ATT}}_{st} = Y_{st}^{(1)} - \widehat{Y}_{st}^{(0)}$. To estimate $\widehat{Y}_{st}^{(0)}$, we employ synthetic control methodology (Abadie and Gardeazabal, 2003; Abadie et al., 2010). We utilize a large control set comprising both COVID-19-related drugs and medications categorized at the GPI-2 level. The inclusion of a rich set of control units allows us to implicitly adjust for multiple confounding factors, including platform-specific effects such as GoodRx market penetration, seasonality, long-term trends, and pandemic-related policies (e.g., shelter-in-place orders, mask mandates, and travel restrictions). We assume these COVID-19-related measures affect all drug claims similarly, except in cases where patients may reasonably forgo a prescription. Notably, by incorporating COVID-19 drugs into the control pool, we also account for shifts in the underlying intensity of the pandemic.

3.2. LNP based regularized SC

As $C \gg n$ here, the standard SC criterion of Abadie et al. (2010) cannot be directly applied, as it may lead to overfitting (Carvalho et al., 2018; Doudchenko and Imbens, 2016), resulting in high prediction error for the estimated ATT. Since we rely on multi-step-ahead ATT estimates for inference, it is essential to ensure adequate control over their predictive accuracy. To address this, Doudchenko and Imbens (2016) and Carvalho et al. (2018) recommend incorporating statistical shrinkage to produce robust SC estimates in high-dimensional settings.

The importance of shrinkage in synthetic control methods has been both theoretically and empirically validated in recent studies (Amjad et al., 2018; Abadie and L'hour, 2021; Ben-Michael et al., 2022, 2021; Kim et al., 2020). In particular, Ben-Michael et al. (2021) demonstrate that shrinkage-based SC estimators can achieve well-controlled predictive risk, and Karmakar et al. (2024) further establish the necessity of shrinkage for robust SC inference in high-dimensional regimes. To this end, we employ a Bayesian Synthetic Control (BSC) method using spike-and-slab shrinkage priors (Ishwaran and Rao, 2005; Malsiner-Walli and Wagner, 2018; Ročková and George, 2018; Antonelli et al., 2019) to estimate state-level ATTs.

We consider modeling Y_{st} as a linear combination of the outcomes of control drugs with Gaussian noise: $Y_{st} = \widehat{Y}_{st}(\boldsymbol{\beta}_s) + \sigma z_{st}$, where $\widehat{Y}_{st}(\boldsymbol{\beta}_s) = \boldsymbol{\beta}_s^\top \boldsymbol{x}_{st} = \sum_{c=1}^{C} \beta_{cs} X_{cst}$, with $\boldsymbol{\beta}_s \in \mathbb{R}^C$ for all s, and z_{st} are i.i.d. standard normal random variables. We consider shrinkage priors on the coefficients $\boldsymbol{\beta}_s = \{\beta_{cs} : c = 1, \ldots, C\}$. Following Thomson et al. (2019) we consider an exchangeable hierarchical structure: for $c = 1, \ldots, C$ and $s = 1, \ldots, S$:

$$\big[\beta_{cs} | \lambda_{cs}\big] \overset{\text{i.i.d.}}{\sim} N(0, \tau\lambda_{cs}^2) \text{ and } \lambda_{cs} \overset{\text{i.i.d.}}{\sim} \texttt{LogitNormal}(m, v)$$

where, $m \in \mathbb{R}$ and $\tau, v > 0$. The above prior structure on $\boldsymbol{\beta}_s$ is equivalent to:

$$\big[\beta_{cs} | z_{cs}\big] \overset{\text{i.i.d.}}{\sim} N\big(0, \tau \exp(2z_{cs})(1 + \exp(z_{cs}))^{-2}\big) \text{ and } z_{cs} \overset{\text{i.i.d.}}{\sim} N(m, v) \ .$$

Note that, the shrinkage of $|\beta_{cs}|$ towards 0 is monotone in z_{cs}. As $z_{cs} \downarrow -\infty$, $\beta_{cs} \to 0$ and $z_{cs} \uparrow \infty$, $\beta_{cs} \to N(0, \tau)$, where, τ is the global shrinkage factor. This Logit-Normal prior (LNP) structure is a global-local continuous shrinkage prior (Bhattacharya et al., 2015; Bhadra et al., 2016; Miller and Harrison, 2018) where τ controls the overall level of shrinkage and the local shrinkage parameters λ_{cs} applies individual shrinkage to each parameter separately. LNP allows for varying levels of shrinkage across different parameters, enabling the model to retain important control units with minimal shrinkage while shrinking less important ones more aggressively (Ročková and George, 2018).

Akin to the horse-shoe priors (Bhadra et al., 2019), LNP is a fully continuous approximation to the mixed spike-and-slab prior which has an atom at origin and a continuous slab (Castillo and Mismer, 2018). Unlike the mixed prior, it is much easier to sample from the LNP based posterior though the hard distinction between zero and non-zero parameters is no longer retained (Malsiner-Walli and Wagner, 2018). However, as LNP substitutes the discrete Bernoulli distribution in the mixed prior with a logit-normal distribution, with appropriate parameter settings it can resemble a U-shaped distribution over $(0, 1)$, concentrating most of its mass near the endpoints. We prefer LNP over the more commonly used Beta distribution (Ročková and George, 2018) as it can be derived as a transformation of standard normal random variables, significantly enhancing the convergence of the sampler (Thomson et al., 2019).

For any fixed σ, τ, m, v, the logarithm of the posterior distribution of $B = \{\boldsymbol{\beta}_s : s = 1, \ldots, S\}$ is given by $\pi(B|Y, X, Z) = \sum_{s=1}^{S} \pi(\boldsymbol{\beta}_s | Y_s, X_s, Z_s)$, where, $\pi(\boldsymbol{\beta}_s | Y_s, X_s, Z_s)$ equals

$$\frac{1}{2\sigma^2} \sum_{t=1}^{n} \left(Y_{st} - \beta_{0s} - \sum_{c=1}^{C} \beta_{cs} X_{cst} \right)^2 + \frac{1}{2\tau} \sum_{c=1}^{C} (1 + \exp(-z_{cs}))^2 \beta_{cs}^2 + \frac{1}{2v} \sum_{c=1}^{C} (z_{cs} - m)^2 \ .$$

We use an inverse-gamma prior on σ, and a half-Cauchy prior on τ. We work with centered variables that directly evaluate β_{0s}. Using code written in the `PyMC 4.0` probabilistic programming language (Salvatier et al., 2016), we sample from the posterior distribution and define the counterfactual estimate as $\widehat{Y}_{st}[\mathtt{m}] := \widehat{Y}_{st}^{(0)}(\widehat{\boldsymbol{\beta}}_s[\mathtt{m}]) = \boldsymbol{x}'_{st}\widehat{\boldsymbol{\beta}}_s[\mathtt{m}]$, where $\widehat{\boldsymbol{\beta}}_s[\mathtt{m}]$ denotes the posterior mean of $\boldsymbol{\beta}_s$. The code for implementing the proposed regularized synthetic control estimates is publicly available from the GitHub repository `https://github.com/rrbhuyan/ivermectin`.

3.3. Risk Properties of RSC under Factor Model structure

The LNP induces sparsity in the SC estimates and controls the prediction error of the ATT estimates at future time-points. We rigorously prove it for a factor model structure (Abadie et al., 2010) on the prescription claims. For any $1 \leq s \leq S$ and $1 \leq t \leq n$, assume the observed IVM prescription claims are generated from a low-dimensional factor model: $Y_{st} = \sum_{k=1}^{K} \phi_{sk}\, \mu_{kst} + \varepsilon_{st}$. Consider counterfactual IVM prescription claims from the same model: $Y_{st}^{(0)} = \sum_{k=1}^{K} \phi_{sk}\, \mu_{kst} + \varepsilon_{st}$ for $t = n+1, \ldots, N$. The claims of the control units for $1 \leq s \leq S$ and $1 \leq t \leq N$ obey: $X_{cst} = \sum_{k=1}^{K} \psi_{csk}\, \mu_{kst} + \varepsilon_{cst}$, where, the coefficient $\boldsymbol{\psi}_{cs} = (\psi_{csk} : 1 \leq k \leq K)$ varies across units and states but is invariant across time whereas the factor $\boldsymbol{\mu}_{st} = (\mu_{kst} : 1 \leq k \leq K)$ is invariant across units but varies across time and state and are bounded by m. The noise terms ε_{cst} are independent with $E(\varepsilon_{cst}) = 0$, $E(\varepsilon_{cst}^2) = \nu^2$ and $E(\varepsilon_{c_1 s_1 t_1} \cdot \varepsilon_{c_2 s_2 t_2}) = 0$ whenever $c_1 \neq c_2$ or $s_1 \neq s_2$ or $t_1 \neq t_2$.

For linear SC predictors of the form $\widehat{Y}_{st}^{(0)}(\boldsymbol{b}) = \boldsymbol{b}'\boldsymbol{x}_{st}$, it follows from appendix B of Abadie et al. (2010) that for any fixed $\boldsymbol{b} \in \mathbb{R}^C$ and for all s and $t > n$, the error of linear SC predictors under factor model structure is upper bounded as:

$$\left|Y_{st}^{(0)} - \widehat{Y}_{st}^{(0)}(\boldsymbol{b})\right| \leq ||\mu_{st}||_2\, \lambda_s \left\{ \mathrm{Im}_s(\boldsymbol{b}) + \nu\rho(\boldsymbol{b})\, \frac{||\boldsymbol{W}_s||_2}{\sqrt{n}} \right\} + \nu\rho(\boldsymbol{b})|\tilde{W}_{st}|, \tag{1}$$

where, $\rho(\boldsymbol{b}) = (1 + ||\boldsymbol{b}||_2^2)^{1/2}$ involves the L_2 norm of $\boldsymbol{b}$, $\mathrm{Im}_s(\boldsymbol{b}) = [n^{-1}\sum_{t=1}^{n}(y_{st} - \boldsymbol{b}'\boldsymbol{x}_{st})^2]^{1/2}$ is the average pre-treatment imbalance (see Theorem 1 of Ben-Michael et al. (2021)) and λ_s is the Kth eigenvalue of $M_s = n^{-1}\sum_{t=1}^{n} \boldsymbol{\mu}_{st}'\boldsymbol{\mu}_{st}$; $\boldsymbol{Z}_s$ is a vector of dimension n with $\boldsymbol{W}_s \stackrel{d}{=} N_n(0, I)$ and $\tilde{Z}_{st}$ is white noise. Also, $\{\boldsymbol{W}_s : s = 1, \ldots, S\}$ and $\{\tilde{W}_{st} : s = 1, \ldots, S; t = n+1, \ldots, N\}$ are independent with each other.

For linear predictors of this form, unless $\|\boldsymbol{b}\|_2$ is properly constrained, the error in (1) can become unbounded. Given that the number of control units is very large and $C \gg n$, the use of unregularized predictors can easily lead to this issue. However, imposing a constraint on $\|\boldsymbol{b}\|_2$ indiscriminately may also result in large imbalance that will be reflected in $\mathrm{Im}_s(\boldsymbol{b})$. To address this, the LNP prior induces sparsity in the state-specific coefficient vectors $\boldsymbol{b}_s$, thereby achieving a balance between controlling the estimation error and maintaining low imbalance.

Next, consider an exact characterization of the difference between Y_{st} and its linear estimator $\widehat{Y}_{st}(\boldsymbol{b}) = \boldsymbol{b}'\boldsymbol{x}_{st}$ for $t = 1, \ldots, N$. This difference can be decomposed into the sum of two distinct components: one that depends on the model parameters and one that does not. Specifically, for any $\boldsymbol{b} \in \mathbb{R}^C$, define the residual $R_{sk}(\boldsymbol{b}) = \phi_{sk} - \sum_{c=1}^{C} b_c \Psi_{csk}$, and let $R_s(\boldsymbol{b}) = \{R_{sk}(\boldsymbol{b}) : k = 1, \ldots, K\}$. The decomposition result is stated below. To simplify the presentation and to state the results uniformly for all $t = 1, \ldots, n$, we omit the superscript on the counterfactual in the display below.

Lemma 1 *Under factor model structure, for any $\boldsymbol{b} \in \mathbb{R}^C$, Y_{st} can be decomposed as:*

$$Y_{st} = \widehat{Y}_{st}(\boldsymbol{b}) + \sum_{k=1}^{K} R_{sk}(\boldsymbol{b})\, \mu_{kst} + \rho(\boldsymbol{b})\, \zeta_{st}, \text{ for all } s \text{ and } t \leq n\,, \tag{2}$$

where, $\rho(\boldsymbol{b}) = \nu\,(1 + \|\boldsymbol{b}\|_2^2)^{1/2}$ and ζ_{st} has standard normal distribution and independent among each other for all (s, t) pairs.

If the linear coefficients $\boldsymbol{b}$ can be chosen such that both residual components in (2) are small, then the counterfactual outcome can be well approximated. Note that our RSC method considers state-specific linear predictors with coefficients $\boldsymbol{\beta}_s$, and ignores the first residual term in (2) as well as the dependence of the scale of the second term on $\|\boldsymbol{\beta}_s\|$. In this sense, it operates on a misspecified version of (2). However, the resulting predictors can still accurately approximate the counterfactual outcomes generated from the factor model. To ensure this, we impose a sparsity assumption on the factor structure. Specifically, we assume that although there is a very large number (C) of control units, for any treated unit s, the coefficient vector $\boldsymbol{\phi}_s = (\phi_{s1}, \ldots, \phi_{sK})'$ can be well approximated by a linear combination $\boldsymbol{\Psi}_s$ involving only ℓ_s control units, where $\ell_s \ll C$ for all s. The set of contributing control units may vary across treated units, and the identity of

these effective control units is unknown a priori. We show that, with high probability, the proposed regularized synthetic control (SC) method based on the LNP (latent noise perturbation) framework accurately selects this subset of relevant control units. Before proceeding, we formally state the sparse factor model assumption. Sparsity assumptions of this type have been widely adopted in the synthetic control literature (see Abadie (2021); Chen and Li (2024) for a comprehensive review).

Assumption 1. We consider an asymptotic regime where $n \to \infty$ and $\liminf_{n\to\infty} C/n > 1$. For each $s \in \{1, \ldots, S\}$, we assume that there exists $\boldsymbol{\beta}_s^* \in \mathbb{R}^C$ such that:
(a) Let $\delta_n := \max_k |R_{sk}(\boldsymbol{\beta}_s^*)|$. Then, $n\delta_n \to 0$ as $n \to \infty$.
(b) Let $\alpha_n := \|\boldsymbol{\beta}_s^*\|_\infty$ and $\ell_n := \|\boldsymbol{\beta}_s^*\|_0$, where $\|\cdot\|_0$ denotes the number of nonzero entries and $\|\cdot\|_\infty$ denotes the supremum norm. Then, $\lim_{n\to\infty} n^{-1}\ell_n(\log C)^3 = 0$ and $\lim_{n\to\infty} \alpha_n/n = 0$.

Note that, since $C \gg n$, ignoring benign pathological cases we have $\inf_{\boldsymbol{b}\in\mathbb{R}^C} R_{sk}(\boldsymbol{b}) = 0$ for all k and s. Assumption 1 guarantees the existence of at least one unknown $\boldsymbol{b}_s^* \in \mathbb{R}^C$ with controlled L_0 and L_∞ norms such that $R_{sk}(\boldsymbol{b}_s^*)$ is asymptotically negligible. Moreover, if ϕ_{sk} and Ψ_{csk} are uniformly bounded and the matrix $\boldsymbol{\psi}_{.s.}$ has full rank, then Assumption 1 automatically holds, as $R_{sk}(\boldsymbol{b}_s^*)$ would be exactly zero.

In the context of (1), Assumption 1 ensures that both the imbalance $\mathrm{Im}_s(\boldsymbol{b}_s^*)$ and the variance $\|\boldsymbol{b}_s^*\|_2$ remain controlled as $n \to \infty$. In such a set-up, we can set the hyper-parameters of LNP prior based on n and C only (and without using any information of the sparsity level) such that its posterior concentrates around $\boldsymbol{\beta}_s^*$ and so, the resultant SC predictor $\widehat{Y}_{st}^{(0)}[\mathtt{m}]$ have controlled predictive risk. To arrive at this result, based on calculations in Chatterjee (2013), we first show that the SC predictor $\widehat{Y}_{st}^{(0)}[\mathtt{mo}] := \widehat{Y}_{st}(\widehat{\boldsymbol{\beta}}_s[\mathtt{mo}]) = \boldsymbol{x}'_{st}\widehat{\boldsymbol{\beta}}_s[\mathtt{mo}]$, based on the posterior mode $\widehat{\boldsymbol{\beta}}_s[\mathtt{mo}]$ of the LNP inherits the slow rate associated with the LASSO predictor (Tibshirani and Wasserman, 2017).

Theorem 2 *Under Assumption 1, as $n \to \infty$, there exists an LNP prior sequence calibrated only on n and C, such that its posterior mode based SC predictions $\widehat{Y}_{st}^{(0)}[\mathtt{mo}]$ satisfy:*

$$E\left(Y_{st}^{(0)} - \widehat{Y}_{st}^{(0)}[\mathtt{mo}]\right)^2 = O\big(\mathcal{R}_n^{(1)} + \mathcal{R}_n^{(2)} + \mathcal{R}_n^{(3)} + \mathcal{R}_n^{(4)}\big)\,,$$

for $1 \le s \le S$ and $n+1 \le t \le N$, where the error terms on the right side are, $\mathcal{R}_n^{(1)} = \|\boldsymbol{\beta}_s^*\|_1^2\,(\log C)^{3/2}\,n^{-1/2}$, $\mathcal{R}_n^{(2)} = \|\boldsymbol{\beta}_s^*\|_1\,\delta_n\,\|\boldsymbol{\mu}_{st}\|_1\,(\log n)^{1/2}$, $\mathcal{R}_n^{(3)} = \delta_n^2\,\|\boldsymbol{\mu}_{st}\|_1^2$ and $\mathcal{R}_n^{(4)} = \nu^2(1 + \|\boldsymbol{\beta}_s^*\|_2^2)$.

The proof of Theorem 2 is provided in the appendix. The first term on the right-hand side of the bound reflects the slow convergence rate of the LASSO predictor. This slower rate is justified, as we make no assumptions regarding the correlations among the control variables, which may be quite high for certain controls. The second and the third term capture the prediction error due to the model misspecification. The last term $\mathcal{R}_n^{(4)}$ is due to future variability and is the unavoidable part of the prediction error (Ben-Michael et al., 2021). Moreover, if the time-varying components $\boldsymbol{\mu}$ in the factor model are uniformly bounded, then under Assumption 1, the first three terms in the bound become asymptotically negligible. Consequently, the prediction error is essentially $\mathcal{R}_n^{(4)}$, which represents the theoretically achievable lower bound using synthetic control methods with perfect oracle knowledge of $\boldsymbol{\beta}_s^*$.

The above result can be extended for the LNP posterior mean based SC estimate $\widehat{Y}_{st}^{(0)}[\mathtt{m}]$. In the appendix, using the results in (Song and Liang, 2023), we show that $\{\widehat{Y}_{st}^{(0)}[\mathtt{m}] : n+1 \le t \le T\}$ concentrates around $\{\boldsymbol{x}'_{st}\boldsymbol{\beta}_s^* : n+1 \le t \le T\}$ with high probability. It provides asymptotic control on the predictive risk of the proposed regularized SC estimates over post treatment time-points. We subsequently use the statewise $\widehat{\mathrm{ATT}}_{st}$ for subsequent analysis.

Theorem 3 *Under Assumption 1, the posterior predictive density induced by the LNP prior sequence in Theorem 2 for predicting the counterfactual based on $\mathcal{D}_n(s) = \{Y_{st}, X_{cst} : 1 \le t \le n, 0 \le c \le C\}$*

concentrates around $\boldsymbol{x}'_{st}\boldsymbol{\beta}^*_s$ *at the following rate:*

$$P\left(\log\left\{\pi\left(\|y-\boldsymbol{x}'_{st}\boldsymbol{\beta}^*_s\| \geq c_1\epsilon_n \,\middle|\, \mathcal{D}_n(s)\right)\right\} \geq -c_2 n\epsilon_n^2\right) \leq \exp\left(-c_3 n\epsilon_n^2\right),$$

where $\epsilon_n = c_4\ell_n^{1/2}(\log C)^{3/2}n^{-1/2}$ *and* c_1, c_2, c_3, c_4 *are positive constants. Consequently, the trimmed posterior mean* $\widehat{Y}^{(0)}_{st}[m]$ *concentrates around the true counterfactual* $Y^{(0)}_{st}$ *as:*

$$P\left(\|\widehat{Y}^{(0)}_{st}[m] - Y^{(0)}_{st}\| \leq \left(c_1\epsilon_n + \delta_n\|\boldsymbol{\mu}_{st}\|_1 + c_5\rho(\boldsymbol{\beta}^*_s)\right)\right) \geq 1 - e^{-c_3 n\epsilon_n^2} \text{ for all } t \geq n.$$

For ease of reference, we summarize the main notations used in this section in Appendix Table 7. While we assume a factor model structure to establish the risk properties of the LNP-based SC method, this assumption is not required for applying the method. As such, under correct specification, the dynamic multilevel latent factor model of Pang et al. (2022) is more flexible and encompasses our setup. However, as the two approaches are based on fundamentally different principles, their relative performance can differ when the factor model assumption is violated; we illustrate this in Sec. 6 through numerical experiments.

3.4. Persistence Analysis of IVM Overconsumption

In addition to addressing the $C \gg n$ problem, the proposed RSC method overcomes several issues of the difference-in-differences model (Imbens and Rubin, 2015). The difference-in-differences approach would require us to pick one control drug that satisfies all the necessary assumptions, including the parallel trend assumption, from the entire set of potential controls or treat all controls as equally important, which significantly reduces the flexibility of the model. An advantage of the proposed LNP based Bayesian RSC approach over frequentist SC methods is straightforward statistical inference, facilitating hypothesis testing.

We use the aggregated state-wise ATT estimates to plot the overall US effects over time as several counter measures were implemented after n. We also look at the effects for different sub-group of the states. Results are robust to an alternative prior, the horseshoe (Kim et al., 2020), although in some instances the horseshoe prior-based model did not converge. The critical and untestable assumption is that the relationship between the focal drug and the control drugs/GPI2 categories in the pre-treatment period would continue to hold after the treatment.

3.5. Analysis of State-Level Countermeasure Effectiveness

We use variation in state-level support for the Republican party in the 2016 election to assess how state-level ATTs vary with political beliefs. Specifically, we consider state-level ATT estimates $\widehat{\text{ATT}}_{st}$ for $s = 1, \ldots, S$ and $t = T_0 + 1, \ldots, T$. We divide the post-treatment period into three intervals and compute the statewise average of the ATT estimates within each period. The first spans Week 51 of 2020 to the end of Week 1 of 2021, immediately following Dr. Pierre Kory's Senate testimony. The second extends from Week 3 to Week 24 of 2021, covering the early-to-mid 2021 period. The third period begins after Week 24 of 2021; at its onset, key studies supporting IVM use were discredited and vaccination rates approached 67%. This temporal segmentation allows us to examine how the impact of key events on IVM prescribing patterns and public perception evolves over time.

4. Results

4.1. IVM Overconsumption patterns over 20 months

We standardized all the indexed claims by the mean and variances till the intervention n. We set treatment date to April 1, 2020, the date when the first study on IVM was published online. In figure 4, we plot the statewise ATT estimates aggregated across the four interesting time intervals reported in figure 3. *Note that, we do not attempt to separately identify the causal impact of individual media or policy events. Instead, the estimated ATT trajectory captures the cumulative deviation in prescription claims following the onset of widespread IVM attention during the pandemic.* Our analysis reveals consistent temporal patterns in IVM prescription claims across states, with notable differences in magnitude between politically diverse regions.

Prior to the emergence of IVM in the COVID-19 pandemic as a potential treatment option, incremental claims per capita were negligible in all states, as expected. However, post-pandemic, we observe a significant increase in claims, with red states exhibiting nearly four times higher rates than blue states. Comparing the ATT plots in figure 4 with the raw claims in figure 3, we see fundamental differences. For example, comparing between the penultimate and last maps in both figures, we see that the ATTs of Texas, Florida and even California are comparatively much pronounced than raw claims maps in figure 3. This differences show the importance of conducting a causal study based on control drugs rather than reporting the differences based on raw IVM claims only. Next, we study these ATTs as counter measures are introduced.

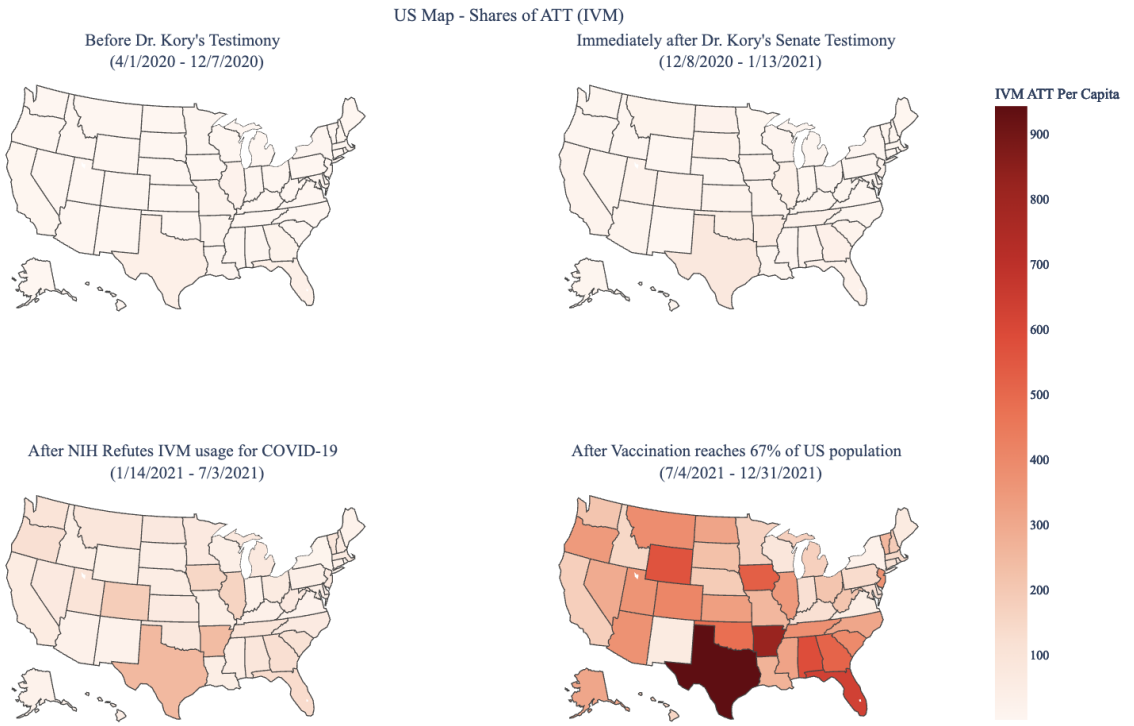

Fig. 4: From top-left row-wise, we have maps of ATT estimates for IVM averaged across the time invervals (a) Event 1 to Event 2 (b) Event 2 to Event 3 (c) Event 3 to Event 6 (d) Event 6 to Event 9. Events listed in Table 1.

In Figure 5, we plot the weekly ATT estimates for IVM prescription claims aggregated across all US states. We see that the two major peaks coincide with the peaks of increased COVID19 case counts in the US. The latter peak is much pronounced than the former peak and occurred after key studies that drew attention to IVM usage for COVID19 was scientifically discredited. Next, we study the fluctuation in the ATTs as countermeasures was introduced.

4.2. Persistence Analysis

Considering the time varying treatment effects in the ATT plot of figure 5 we observe a small, short-lived increase in prescription claims in relation to the counterfactual on April 1, 2020, but the major increase does not occur till December 8, 2020, when Dr. Kory testified for the senate committee. This increase sustains till the end of 2021, with a substantial increase in 2021 coincident with an increase in COVID cases. Thus, we observe a persistent effect of IVM overconsumption, despite federal informative countermeasures.

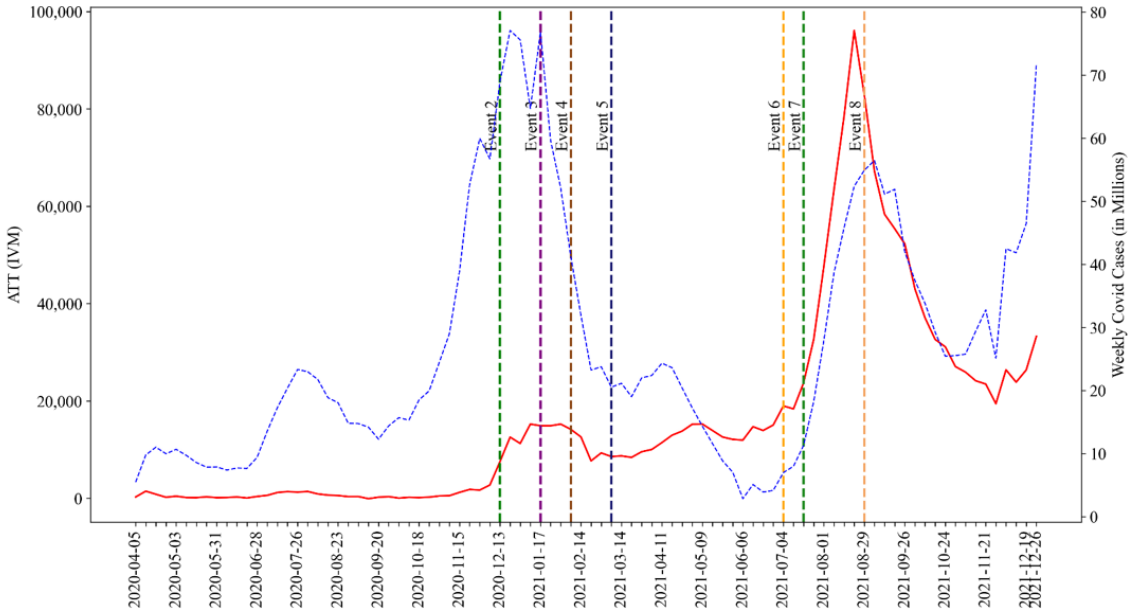

Fig. 5: Plot of weekly ATT estimates for IVM prescription claims aggregated across all US states (in red) and of total US weekly Covid case counts (in blue). The time points marked by dotted vertical lines correspond to major related events listed in Table 1.

Unlike restrictive countermeasures that involves banning consumption, informative countermeasures are actions designed to reduce the spread and impact of misleading information by enhancing public knowledge, promoting critical thinking, and improving access to accurate information. All IVM countermeasures were informative in nature.

Table 2. Change in Incremental IVM Claims Over Time

Interval	Significance	ATT Increase
12/08/2020 – 01/13/2021	Immediately after Dr. Kory's Senate Testimony	6.15 times
01/14/2021 – 07/03/2021	After NIH Refutes IVM usage for COVID-19	4.52 times
07/04/2021 – 12/31/2021	After Vaccination reaches 67% of US population	16.18 times

In table 2, we summarize the information in the weekly ATTs plots of figure 5 by considering three interesting time periods from the span of the study. Note that, as these ATT estimates are based on indexed claims, the estimates unless compared to a base is not interpretative. For each of the time-period, we report the average increase in total US ATTs as relative change over the counterfactual, i.e., for any time-interval $\mathcal{T}$, we report $(\sum_{s\in\mathcal{S},t\in\mathcal{T}} Y_{st}(1) - \hat{Y}_{st}(0))/(\sum_{s\in\mathcal{S},t\in\mathcal{T}} \hat{Y}_{st}(0))$. For approximately a month after Dr Kory's testimony, the relative increase in IVM claims were around 6 times. However, we observe that the highest increase on average which was 16 times occurred after 67% of the US population had received at least one dose of the COVID19 vaccine. Also, at that point there has been scientific evidence refuting the initial studies that suggested IVM usage for COVID19. This shows that the effect of overconsumption persisted even after counter measures and vaccination.

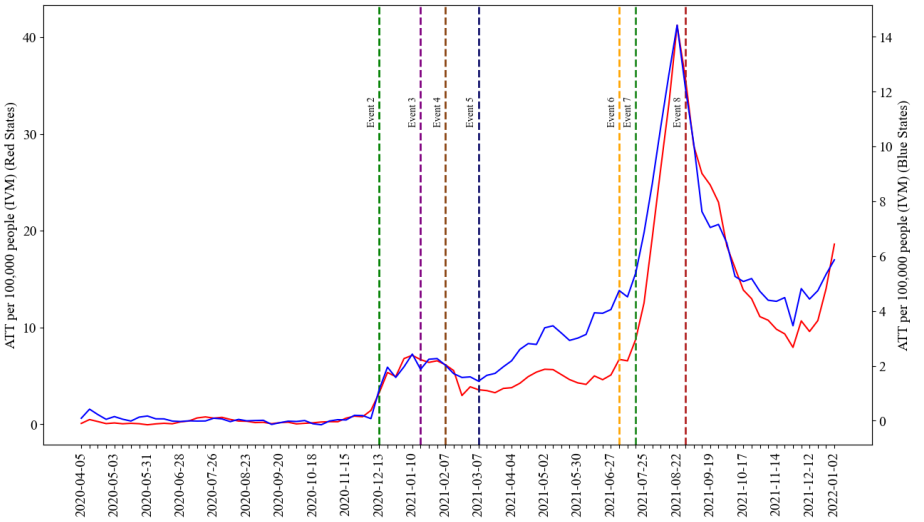

Fig. 6: Plot of weekly ATT estimates for IVM prescription claims aggregated across Blue and Red states. The time points marked by dotted vertical lines correspond to major related events listed in Table 1.

4.3. State-wise variability in regulatory countermeasure efficacy

We next compute weekly ATT estimates per capita for Blue and Red states. For each group, we aggregate ATT estimates across states and normalize by the corresponding total population. The resulting weekly time series is shown in Figure 6, where we observe noticeable differences between the two groups. *To further examine these differences and their relationship with political affiliation, we partition the study period into three intervals and present scatter plots of state-level average ATT estimates against the conservativeness index. Figure 7 plots ATT per million against the conservativeness index for each state across these periods.*

In the period immediately following Dr. Kory's Senate testimony , we observe a modest positive association, with Texas as a notable outlier. This association largely disappears in the early-to-mid 2021 period, suggesting that IVM overconsumption was broadly distributed across states, consistent with widespread uncertainty regarding COVID-19 treatment options at the time. In contrast, the association strengthens markedly in late 2021, after vaccines became widely available and key IVM studies were discredited. This pattern indicates that federal countermeasures were less effective in more politically conservative states once clearer scientific evidence had emerged.

5. Robustness Checks and Additional Analyses

5.1. Convergence Diagnostics of our method

Figure 8 summarizes the potential scale reduction factor ($\hat{R}$) diagnostics (Gelman and Rubin, 1992) for all 50 statewide models. For each state, we report the minimum and maximum $\hat{R}$ values, which provides a compact view of the overall convergence behavior of each fitted model. In MCMC convergence diagnostics, $\hat{R}$ values less than 1.01 indicate excellent mixing across chains. Across the 50 models in figure 8, most $\hat{R}$ values are very close to 1, suggesting good convergence. A small number of fits exhibit $\hat{R}$ values above 1.005, but none of them exceed the 1.01 threshold, indicating that convergence is good though not uniformly perfect across all state-level fits.

5.2. Placebo Tests

As a falsification test, we assume that the treatment occurred at the beginning of 2020. Specifically, we shift the treatment date back by three months from April 1, 2020, the date when the first study suggesting

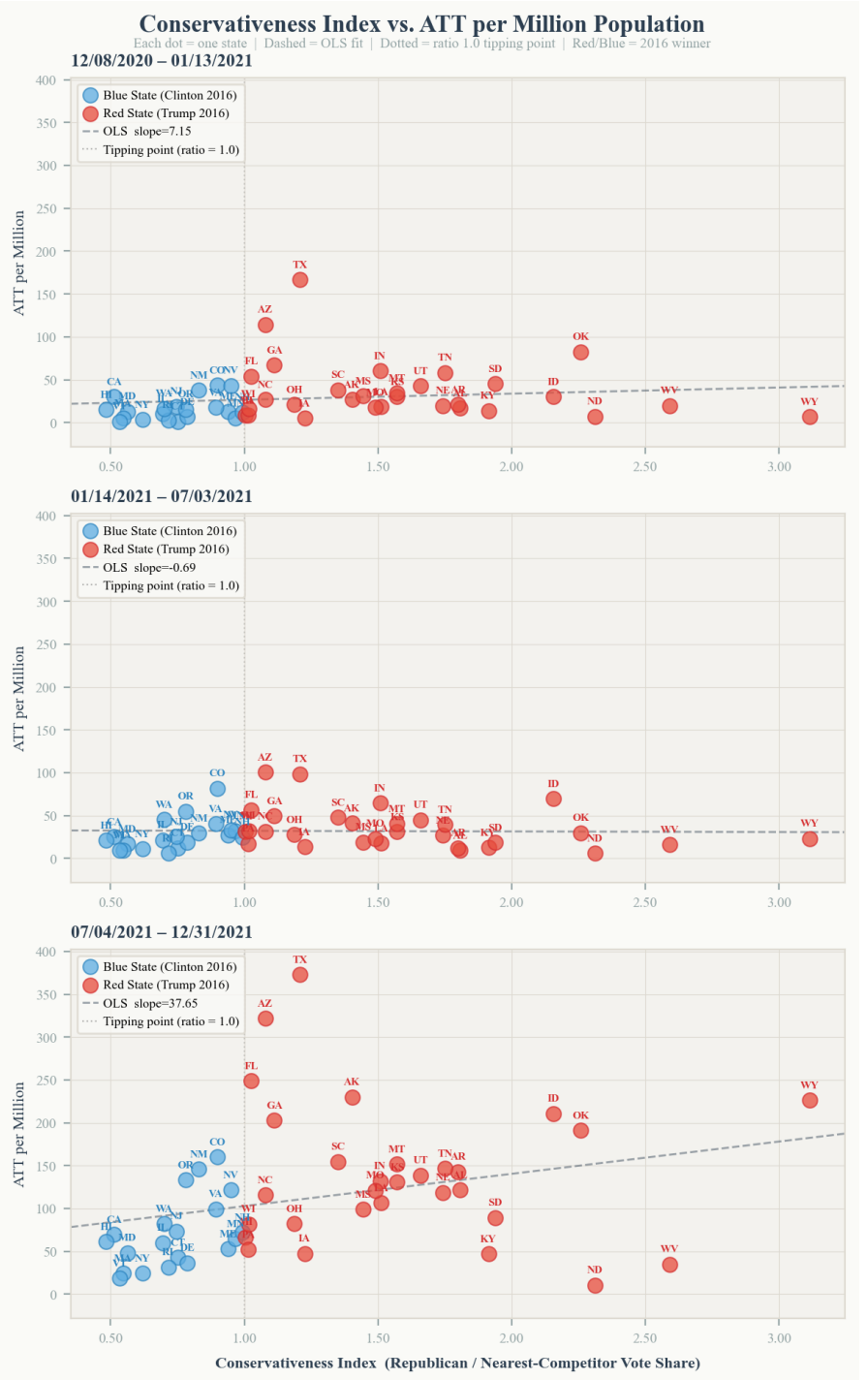

Fig. 7: *Scatterplot of per capita ATT estimates versus the conservativeness index across three study periods.*

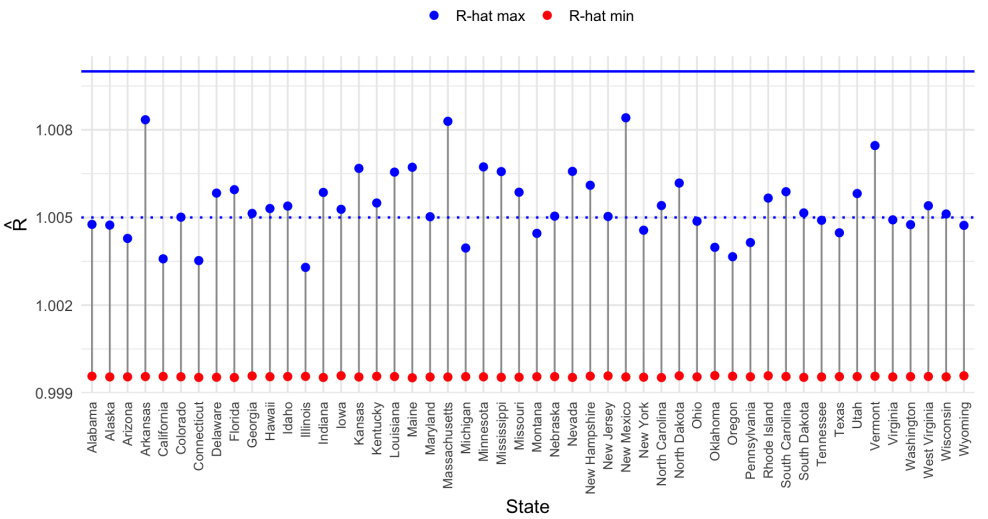

Fig. 8: *Potential scale reduction factor ($\hat{R}$) convergence diagnostics across the 50 statewide models. For each state, the red and blue points indicate the minimum and maximum $\hat{R}$ values, respectively. The horizontal bold line marks the threshold of* 1.01*, below which $\hat{R}$ values indicate excellent convergence of the MCMC chains.*

that IVM might be effective was published, to January 12, 2020. We then re-estimate the counterfactual outcomes using our proposed method. *Figure 9 displays the observed pre-treatment time series (in red) for the weekly U.S. IVM claims together with the corresponding fitted counterfactual values from the model (in blue), along with the associated 95% credible interval.* Focusing on the 12-week placebo window following January 12, 2020, the observed series remains within the uncertainty band throughout the period, indicating that the estimated placebo effects are not statistically distinguishable from zero. This falsification exercise provides evidence of adequate model fit in the absence of treatment and helps rule out spurious inference arising from overfitting or model misspecification. The results also suggest that there was no evidence of stockpiling in anticipation of potential shortages prior to April 1, 2020.

5.3. Consistency of primary findings across related causal inference methods

We apply three related Bayesian causal inference methods to our data. Specifically, we consider two Bayesian synthetic control approaches, `bayeslasso` *and* `horseshoe`*, which impose Laplace (Park and Casella, 2008) and horseshoe priors (Bhadra et al., 2016), respectively, to shrink the control coefficients. In addition, we consider the dynamic multilevel latent factor model–based causal estimation method* `plx` *proposed by Pang et al. (2022), which provides a Bayesian alternative to the synthetic control framework. To implement* `bayeslasso` *and* `horseshoe`*, we used Stan through the* `BayesReg` *package with its default settings. For* `plx`*, we used the code provided in Pang et al. (2021), and the hyper-parameter settings were kept exactly as specified in the original code.*

We begin with the placebo (falsification) test described in the previous subsection, applied to the weekly U.S. IVM claims series with the intervention date shifted to January 12, 2020. For each estimator, we estimate the counterfactual trajectory and compute post-placebo ATT estimates over the subsequent 12 weeks. In the absence of anticipatory effects and under correct model specification, these placebo ATTs should remain close to zero; accordingly, the corresponding credible intervals should include zero for most (ideally all) weeks within the placebo window. Figure 10 summarizes the placebo ATT estimates for the competing methods. The placebo series from all four methods remain centered around zero over the 12-week window, and their uncertainty bands contain zero throughout, indicating that each method passes the falsification test. At the

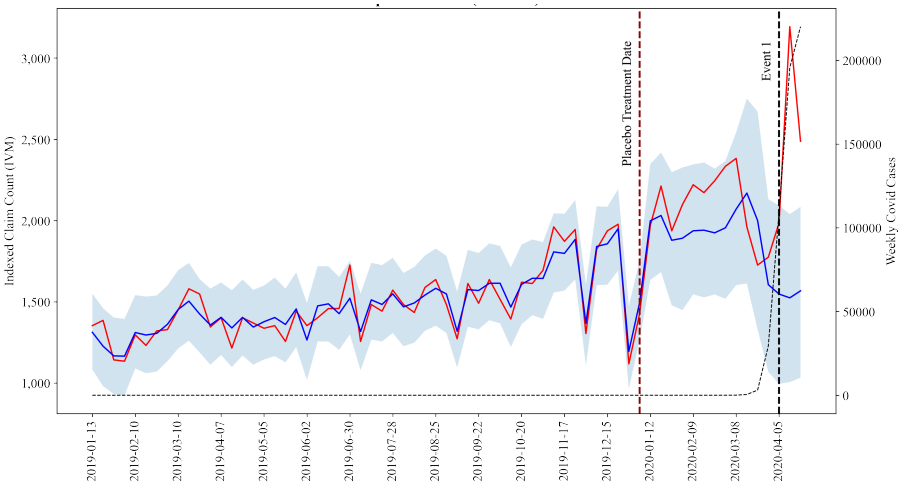

Fig. 9: *Placebo test with the treatment date shifted to January 12, 2020 for the weekly U.S. ivermectin (IVM) claims time series. The red curve shows the observed outcome, while the blue curve shows the fitted trajectory. The shaded region denotes the 95% credible interval for the counterfactual prediction.*

same time, the credible intervals for `plx` *are noticeably wider, reflecting lower precision. The intervals for* `horseshoe`*,* `bayeslasso`*, and our method are considerably tighter, with our method exhibiting the narrowest uncertainty bands among the four. Table 3 quantifies these findings by reporting, for each method, the average ATT across the 12 post-placebo weeks together with the average length of the 95% credible interval.*

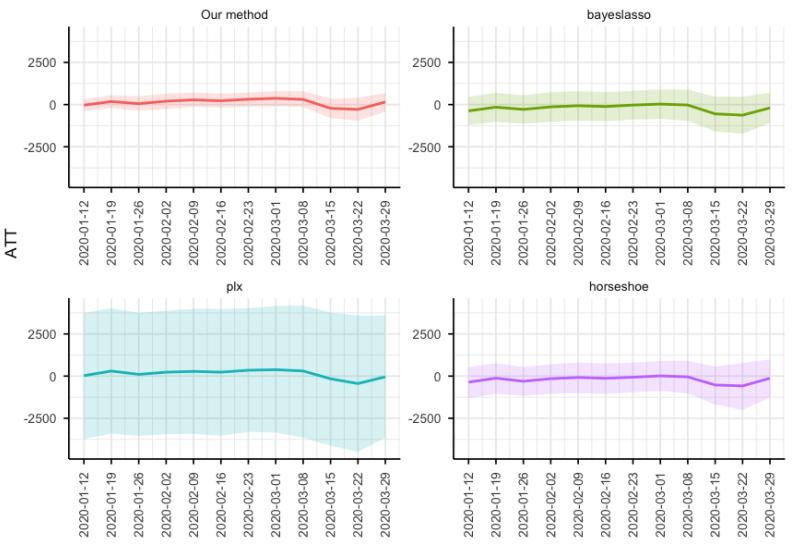

Fig. 10: *Placebo falsification test on U.S. IVM claims with shifted treatment date (January 12, 2020). Lines denote weekly ATT estimates; shaded regions denote the associated 95% credible intervals.*

Table 3. *Average ATT estimates, average 95% credible interval lengths, and false-positive rates across the 12-week placebo period for the different methods.*

Method	Avg. ATT	Average Length
Our method	129.38	929.36
`bayeslasso`	-209.93	1801.54
`plx`	130.15	7526.28
`horseshoe`	-207.23	1950.78

We next apply these methods to the U.S. weekly IVM claims series using the correct intervention date of April 1, 2020. Table 4 summarizes the estimated ATTs and the corresponding average widths of the 95% credible intervals for each method over the treatment period, partitioned into four sub-periods. For each estimator and sub-period, we report (i) the mean of the weekly ATT point estimates and (ii) the mean width of the associated 95% credible interval. Across all methods, the direction of the estimated ATTs is broadly consistent within each sub-period, supporting the robustness of the substantive findings to the choice of estimator. However, the average interval widths reveal notable differences in uncertainty quantification. Our method produces stable ATT estimates across sub-periods while maintaining relatively compact uncertainty intervals, suggesting that the detected treatment effects are estimated with good precision. In contrast, `plx` *produces substantially wider credible intervals in every sub-period, indicating considerably lower precision despite ATT point estimates of similar direction and magnitude.*

Table 4. *ATT estimates and average length of the 95% credible intervals for the four treatment-period events across methods.*

Event	Method	Avg. ATT	Avg. Interval Length
Event 1 – Event 2 (04/01/2020 – 12/07/2020)	Our method	3155.07	1083.14
	`bayeslasso`	3088.72	1625.67
	`plx`	3393.40	7175.58
	`horseshoe`	3257.35	1613.74
Event 2 – Event 3 (12/08/2020 – 01/13/2021)	Our method	9001.85	1557.02
	`bayeslasso`	8889.76	2243.30
	`plx`	9450.30	7307.85
	`horseshoe`	8896.49	2376.71
Event 3 – Event 6 (01/14/2021 – 07/03/2021)	Our method	34722.35	1824.49
	`bayeslasso`	34735.13	2174.54
	`plx`	35196.12	7421.71
	`horseshoe`	34785.89	2190.49
Event 6 – Event 9 (07/04/2021 – 12/31/2021)	Our method	29270.68	1982.01
	`bayeslasso`	29415.90	1822.02
	`plx`	29628.10	7473.16
	`horseshoe`	29483.23	1889.74

6. Comparative Study Using Synthetic Data

We conduct a simulation study using synthetic data to compare estimation accuracy and uncertainty quantification across methods. The simulated panel contains 200 *total units, consisting of* 1 *treated unit and* 199 *control units. The data-generating process is designed to mimic key structural features of the IVM dataset. In the IVM data, the first principal component explains* 75.8% *of the variance, and the autocorrelation of this component is* 0.87*. Guided by these features, we generate a synthetic donor matrix* X *with the control units from a latent factor model driven by* 10 *latent variables. The first factor is constructed to dominate the remaining factors so that the resulting donor panel exhibits a similar principal component*

structure to the real data (the first component explains approximately 75% of the variance). The treated unit's untreated trajectory $Y^{(0)}$ is generated from the same latent environment.

After generating the untreated outcome path, we introduce treatment through a post-treatment ATT. The observed treated outcome equals the untreated trajectory in the pre-treatment period and equals the untreated trajectory plus a time-varying ATT after treatment, i.e., $Y_t = Y_t^{(0)}$ for $t \leq T_0$ and $Y_t = Y_t^{(0)} + \mathrm{ATT}_{t-T_0}$ for $t > T_0$. The ATT increases smoothly over time rather than appearing as an instantaneous jump, reflecting gradually accumulating effects. Specifically, the effect follows $\mathrm{ATT}_j = \mathrm{ATT}_{\max}(1 - \exp(-j/\tau))$ for $j = 1, \ldots, T$ with $\tau = 10$, reaching approximately 63% of $\mathrm{ATT}_{\max}$ at $j = 10$ and 95% by $j = 30$.

Table 5 summarizes the performance of the four methods on the synthetic data. Estimation accuracy is measured by RMSE and bias, while coverage reports the proportion of post-treatment time points for which the true ATT lies within the 95% credible interval; CI width denotes the average length of the 95% credible interval. `plx` achieves the highest coverage but at the cost of very wide 95% credible intervals and substantially poorer estimation accuracy. In contrast, `horseshoe` and `bayeslasso` attain the lowest RMSE and small bias with much narrower intervals, but their coverage drops to 78.5%. Our method provides a more balanced performance, achieving competitive RMSE and near-zero bias while maintaining higher coverage (90.8%) than the other Bayesian SC methods, with 95% credible intervals that remain far tighter than those of `plx` (average width 1.12 versus 4.00).

Table 5. *Simulation performance comparison. RMSE and bias are reported as mean ± s.d. across replications. Coverage denotes empirical 95% credible interval coverage and CI width denotes the average 95% credible interval length.*

	Our method	`bayeslasso`	`plx`	`horseshoe`
RMSE	0.325 ± 0.039	0.278 ± 0.028	0.847 ± 0.574	0.274 ± 0.027
Bias	0.004 ± 0.069	0.002 ± 0.046	-0.043 ± 0.439	0.003 ± 0.042
Coverage	90.8%	78.5%	92.5%	73.7%
CI width	1.12	0.70	4.00	0.63

7. Discussion

This study provides a causal analysis of the medium-term effects of ivermectin (IVM) overuse and the effectiveness of federal informational countermeasures during the COVID-19 pandemic. Our findings show that IVM overuse persisted well beyond the initial reports of its potential therapeutic benefits, despite regulatory advisories and public health messaging. Although these countermeasures had some effect, their overall impact was limited and substantial residual overuse remained.

We also find significant heterogeneity in the effectiveness of countermeasures across states. States with higher levels of political conservativeness experienced greater per capita overconsumption and exhibited lower responsiveness to federal advisories. Moreover, the release of definitive scientific evidence discrediting IVM's efficacy had only a modest effect on reducing its use.

These results highlight the limitations of federal informational countermeasures during the pandemic and suggest the need for more targeted health communication strategies. The large variation in state-level responses indicates that uniform messaging may be insufficient. Communication strategies tailored to specific populations—taking into account behavioral, cultural, and political contexts—may improve effectiveness. Incorporating data analytics and behavioral insights into communication design could further enhance message uptake and public compliance. Addressing these challenges will require coordinated efforts across public policy, healthcare, behavioral science, and communication research to develop more adaptive responses to future public health crises.

Finally, the Blue and Red state patterns in figure 6 reveal notable temporal dynamics. The two series track each other closely (in shape) until Event 4, diverge by Event 7, and begin to converge again before Event 8. Understanding the mechanisms behind this divergence and convergence is an important direction for future research because it may shed light on how information shocks, regulatory actions, and behavioral responses jointly influence drug utilization.

A. Comparison with Related Empirical Studies

Two recent studies are closely related to our work on Ivermectin prescribing during the COVID-19 pandemic: Rockwell et al. (2025) and Barnett et al. (2022). While these papers provide valuable descriptive evidence, our study differs from them in several important respects. We outline these differences below and summarize them in Table 6.

Comparison with Rockwell et al. (2025):

- ***Causal vs. descriptive analysis.*** *Rockwell et al. (2025) primarily report descriptive utilization trends based on comparisons between observed and expected prescribing rates. In contrast, our study adopts a synthetic control framework to estimate state-specific causal effects (ATTs) by constructing explicit counterfactual prescribing trajectories.*
- ***Political heterogeneity.*** *Rockwell et al. (2025) do not examine heterogeneity associated with political affiliation. Our analysis investigates whether prescribing dynamics differ across states with varying political characteristics.*
- ***State-level granularity.*** *Our framework produces 50 state-specific ATT trajectories over a 72-week period, allowing us to study substantial geographic variation. In contrast, Rockwell et al. (2025) focus primarily on national aggregates.*

Comparison with Barnett et al. (2022):

- ***Temporal coverage.*** *Barnett et al. (2022) analyze the period January–December 2020, which precedes the large surge in Ivermectin prescribing observed in 2021. Our dataset spans 42 months and covers the full duration of the public health emergency.*
- ***Primary research focus.*** *Barnett et al. (2022) study county-level associations between political voting patterns and prescribing behavior using a difference-in-differences framework. Our study instead focuses on quantifying the persistence of Ivermectin overconsumption following multiple countermeasures and examining heterogeneity in treatment effects across states.*
- ***Methodological framework.*** *Our analysis employs a synthetic control approach using 173 control medications to construct counterfactual prescribing trajectories and estimate causal ATTs for each state. In contrast, Barnett et al. (2022) use a difference-in-differences design comparing prescribing trends across political quartiles and rely on a small set of reference medications.*

Table 6. Comparison of study designs and contributions

Dimension	*Barnett et al. (2022)*	*Our Study*	*Rockwell et al. (2025)*
Causal strategy	*Comparative DiD (political quartiles)*	*Regularized synthetic control (state-specific)*	*Descriptive analyses (no causal estimator)*
Identification	*Differential prescribing by political quartile*	*Large control set (173 drugs) used to construct counterfactuals*	*2019 baseline used as reference for expected rates*
Study period	*Jan 2019–Dec 2020 (24 months)*	*Jan 2019–Dec 2021 (36 months)*	*Jan 2020–Jun 2023 (42 months)*
Persistence analysis	*No*	*Yes*	*Yes*
Political heterogeneity	*Yes (county GOP vote share)*	*Yes (state-wise index for conservativeness)*	*No*

B. Further Robustness Checks

Our robustness checks broadly fall into two types: data/model related and alternative explanations for our observed effects.

Single Platform Data. Our data is sourced from a single platform, GoodRx. We view our analysis as an index of claim patterns in the market. As a data check, IVM claims at the national level from another data supplier, who wishes to be anonymous. The anonymous data supplier is a hub for an extensive pharmacy network, routing real-time Rx claims transactions between pharmacies and pharmacy benefit managers (PBMs). This connectivity offers broad coverage of pharmacies, prescribing providers, and PBMs sourced as a part of the business platform. Coverage includes specialty pharmacy claims, general distribution prescription drugs, and durable medical equipment claims filled via pharmacy. The data supply is 1st party and deliverable on a t+1 basis from the claims submission date. The Pearson correlation between GoodRx data and the data supplier data for IVM claims is 0.9.

Choice of Controls. We estimated models with other drugs that were used to treat Covid-19 and available at pharmacies along with aggregate GPI2 categories as controls. We identified drugs used for Covid-19 treatment from the NIH's treatment advisories between April 21, 2020 and December 30,2021 (inclusive) for Covid-19. We check if the results substantively remain the same if we use different control sets. We tried (a) top 200 drugs by volume (b) use a random sampling of 200 drugs, and (c) use top 100 drugs by volume and aggregate GPI2 categories as controls. We see very similar results in all the three cases. To avoid repetition, we present the results from case (c) only. We plot the ATT estimates based in case (c) in figure 11. We see that with this alternative specification the treatment effect estimates are even larger.

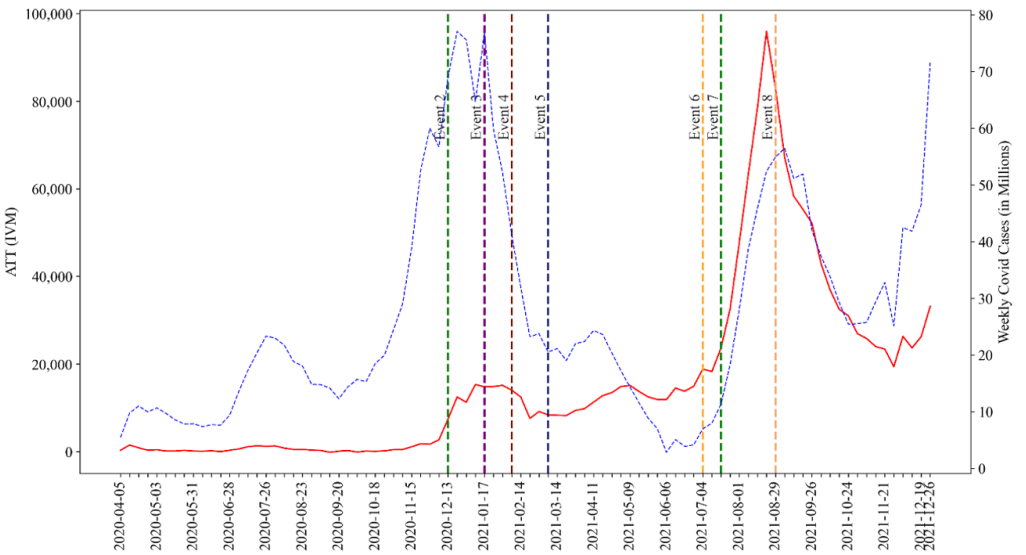

Fig. 11: Based on an alternative control set we plot the ATT estimates for IVM prescription claims aggregated across all US states (in red) and of total US weekly Covid case counts (in blue).

Covid Cases. Inclusion of covid death rates in place of case rates does not substantially alter our results.

Differential Media Coverage of Countermeasures across states. It is quite possible that the media covered countermeasures differently across states. This could mitigate the effect of countermeasures. However, our study focuses on prescription claims - where physicians have to sign off on the prescription. Ideally, experts like physicians should be able to directly access updated information outside of media sources.

Substitutes for IVM. There are no over-the-counter (OTC) substitutes for IVM and IVM purchase requires a prescription. Therefore, differential access to over-the-counter substitutes does not explain our results. Some consumers used animal formulations of IVM, but this is not pertinent to our study as such consumption does not include physician decisions.

C. Proofs of the results in Section 3.1

C.1. Proof of Lemma 1

Based on our factor model, we have $Y_{st} = \sum_{k=1}^{K} \phi_{sk}\, \mu_{kst} + \varepsilon_{st}$, and similarly the controls are $X_{cst} = \sum_{k=1}^{K} \psi_{csk}\, \mu_{kst} + \varepsilon_{cst}$. The residual loadings for any $\boldsymbol{b} \in \mathbb{R}^c$ is defined as

$$R_{sk}(\boldsymbol{b}) \;=\; \phi_{sk} \;-\; \sum_{c=1}^{C} b_c\, \psi_{csk}.$$

Since $\phi_{sk} = R_{sk}(\boldsymbol{b}) + \sum_{c=1}^{C} b_c \psi_{csk}$, we can rewrite Y_{st} as

$$\begin{aligned} Y_{st} &= \sum_{k=1}^{K} \Big(R_{sk}(\boldsymbol{b}) + \sum_{c=1}^{C} b_c \psi_{csk} \Big) \mu_{skt} + \varepsilon_{st} \\ &= \sum_{k=1}^{K} R_{sk}(\boldsymbol{b})\, \mu_{kst} + \sum_{c=1}^{C} b_c \sum_{k=1}^{K} \psi_{csk} \mu_{kst} + \varepsilon_{st}\,. \end{aligned}$$

Next, by using the factor models for the control units, we can replace $\sum_{k=1}^{K} \psi_{csk} \mu_{kst}$ with $X_{cst} - \varepsilon_{cst}$, to simplify as follows,

$$\begin{aligned} Y_{st} &= \sum_{k=1}^{K} R_{sk}(\boldsymbol{b})\, \mu_{skt} + \sum_{c=1}^{C} b_c \big(X_{cst} - \varepsilon_{cst} \big) + \varepsilon_{st} \\ &= \sum_{k=1}^{K} R_{sk}(\boldsymbol{b})\, \mu_{kst} + \widehat{Y}_{st}(\boldsymbol{b}) - \sum_{c=1}^{C} b_c\, \varepsilon_{cst} + \varepsilon_{st}\,. \end{aligned}$$

The final result follows by noting that $\varepsilon_{st} - \sum_{c=1}^{C} b_c\, \varepsilon_{cst} \overset{d}{=} \rho(\boldsymbol{b}) \zeta_{st}$, where $\rho(\boldsymbol{b}) = \nu(1 + \|\boldsymbol{b}\|_2^2)^{1/2}$ and ζ_{st} is standard normal.

C.2. Proof of Theorem 2

Consider the setup for a particular state s. We remove subscripts pertaining to state s for ease of notation. In this case we model the pre-treatment IVM claims $\{Y_t : t = 1, \cdots, n\}$, with the misspecified model for a single state as

$$Y_t = \sum_{c=1}^{C} \beta_c X_{ct} + \epsilon_t\,; \quad \big[\beta_c | \lambda_c\big] \overset{\text{i.i.d.}}{\sim} N(0, \tau \lambda_c^2) \text{ and } \lambda_c \overset{\text{i.i.d.}}{\sim} \texttt{LogitNormal}(m, v),$$

For fixed σ, τ and a fixed sequence of λ_c, the negative log likelihood of posterior distribution of β_c without the prior structure on λ_c is

$$l(\boldsymbol{\beta} | Y, X, \lambda) \propto \frac{1}{2\sigma^2} \sum_{t=1}^{n} \big(Y_t - \sum_{c=1}^{C} \beta_c X_{ct} \big)^2 + \frac{1}{2\tau} \sum_{c=1}^{C} \frac{\beta_c^2}{\lambda_c^2}\,.$$

The posterior mode $\widehat{\boldsymbol{\beta}}$ is the minimizer of the above loss function (negative log-likelihood), $\widehat{\boldsymbol{\beta}} = \arg\min l(\boldsymbol{\beta} | Y, X, \lambda)$. The optimization problem of minimizing the loss can be rewritten as a constrained problem for some K as

$$\min_{\beta} \quad \frac{\tau}{\sigma^2} \sum_{t=1}^{n} \big(Y_t - \sum_{c=1}^{C} \beta_c X_{ct} \big)^2 \qquad \text{s.t.} \quad \sum_{c=1}^{C} \frac{\beta_c^2}{\lambda_c^2} \le K\,. \tag{3}$$

We next provide the construction to choose prior hyperparamaters for λ_c.

Lemma 4 *Let $X \sim \texttt{LogitNormal}(-tf(t), t^2)$, where $f(t)$ is the solution to the equation*

$$\frac{f(t)^2}{2} + \log f(t) = (1+u)\log t\,,$$

then as $t \to \infty$, for some $u > 0$, there exists constants k_1, k_2 such that

$$P[X \in (0, t^{-1/2})] = 1 - k_1 t^{-(1+u)} \qquad P[X \in (1 - t^{-1/2}, 1)] = k_2 t^{-(1+u)}$$

Using the above lemma, we set the prior mean $-Cf(C)$ and variance C^2 for λ_c. The limiting distribution closely resembles a Bernoulli distribution. With this choice of parameters $\lambda_c \leq C^{-1/2}$ with probability $1-k_1C^{-(1+u)}$ and the entire sequence of λ_c is bounded by $1/\sqrt{C}$ with probability $(1-k_1/C^{1+u})^C \approx 1-C^{-u}$ for some $u > 0$. Thus, $\lambda_c \geq 1/\sqrt{C}$ for only constants s λ_c's as C grows. Without loss of generality assume that $\lambda_1, \lambda_2, \cdots, \lambda_s$ are greater than $1/\sqrt{C}$ and $\lambda_{s+1}, \lambda_{s+2}, \cdots, \lambda_C \leq 1/\sqrt{C}$. Consider $2s\sqrt{\sum_{c=1}^C \beta_c^2/\lambda_c^2}$,

$$2s\sqrt{\sum_{c=1}^{C} \beta_c^2/\lambda_c^2} = 2s\sqrt{\sum_{c=1}^{s} \beta_c^2/\lambda_c^2 + \sum_{c=s+1}^{C} \beta_c^2/\lambda_c^2} \geq s\sqrt{\sum_{c=1}^{s} \beta_c^2/\lambda_c^2} + s\sqrt{\sum_{c=s+1}^{C} \beta_c^2/\lambda_c^2}$$
$$\geq s\sqrt{\sum_{c=1}^{s} \beta_c^2} + \sqrt{C\sum_{c=s+1}^{C} \beta_c^2} \geq \|\boldsymbol{\beta}\|_1\,.$$

This means that $\sum_{c=1}^C \beta_c^2/\lambda_c^2 \leq K$ implies that $\|\boldsymbol{\beta}\|_1 \leq 2s\sqrt{K}$. Thus consider the following optimization problem instead of (3),

$$\min_{\beta} \sum_{t=1}^{n} \frac{\tau}{\sigma^2}\big(Y_t - \sum_{c=1}^{C} \beta_c X_{ct}\big)^2 \qquad \text{s.t.} \quad \|\boldsymbol{\beta}\|_1 \leq 2s\sqrt{K}\,, \tag{4}$$

where, $\tilde{Y}_t = \sqrt{\tau/\sigma^2}Y_t$ and $\tilde{X}_{ct} = \sqrt{\tau/\sigma^2}X_{ct}$, the problem is thus normalized for the prior parameters. While the objective function of the optimization problems in both (3) and (4) are the equivalent, since $\sum_{c=1}^C \beta_c^2/\lambda_c^2 \leq K$ implies that $\|\boldsymbol{\beta}\|_1 \leq 2s\sqrt{K}$, thus, the feasible set of (4) is a subset of the feasible set of (3). Minimising over a smaller region can only increase (or leave unchanged) the optimal value, so the infimum attained in (4) is at least as large as that for (3). Because our ultimate goal is to derive an upper bound on the prediction error, we concentrate on solving the tighter problem (4). The optimization problem in (4) resembles the problem defined in Chatterjee (2013) to show the bound on the prediction error. We provide a version of the result from Chatterjee (2013), which we use to provide our result

Result (Chatterjee (2013)) *Consider the regression setup $Y_t = \sum_{c=1}^C \beta_c^* X_{ct} + \epsilon_t$ where $\|\boldsymbol{\beta}^*\|_1 \leq K_\beta$ and the errors have variance σ^2. Let MSPE stand for the mean squared prediction error, defined as $E\left(\tilde{Y} - \widehat{Y}\right)^2$ where $\widehat{Y}$ is the prediction based on $\widehat{\boldsymbol{\beta}}$ (restricted to $\|\widehat{\boldsymbol{\beta}}\|_1 \leq K$) and $\tilde{Y}$ is the best possible prediction based on the true parameter $\boldsymbol{\beta}^*$, then*

$$E\left(\tilde{Y} - \widehat{Y}\right)^2 = \tilde{K}M\sigma\sqrt{\frac{\log(p)}{n}} + 2\tilde{K}^2M^2\sqrt{\frac{\log(p)}{n}}\,,$$

where $\max|X_{ct}| \leq M$ and $\tilde{K} := (K + K_\beta)^2$.

In our case, as described in Lemma 1, we have

$$Y_t = \sum_{c=1}^{C} \beta_c^* X_{ct} + \sum_{k=1}^{K} R_k(\boldsymbol{\beta}^*)\,\mu_{kt} + \rho(\boldsymbol{\beta}^*)\,\zeta_t\,. \tag{5}$$

Since we have formulated our optimization problem in (4) using the normalized $\tilde{Y}_t$, the variance in our setting is thus $\tau\rho(\boldsymbol{\beta}^*)/\sigma^2$. Also M is the maximum of n of X_{ct}'s which is of the order $\sqrt{\log n}$. The model

misspecification term $\sum_{k=1}^{K} R_k(\boldsymbol{\beta}^*)\,\mu_{kt}$ also adds another error of $M\tilde{K}\sum_{k=1}^{K} R_k(\boldsymbol{\beta}^*)\,\mu_{kt}$ into our exisiting error. Combining all, we get for out case, given fixed τ and σ,

$$E\left[\left(\tilde{Y}_t^{(0)} - \widehat{Y}_t\right)^2 \Big|\tau,\sigma\right] = \tilde{K}M\sqrt{\frac{\tau\rho(\boldsymbol{\beta}^*)\log C}{\sigma^2 n}} + 2\tilde{K}^2M^2\sqrt{\frac{\log C}{n}} + M\tilde{K}\sum_{k=1}^{K} R_k(\boldsymbol{\beta}^*)\,\mu_{kt}$$
$$= \tilde{K}M\left(\sqrt{\frac{\tau\rho(\boldsymbol{\beta}^*)\log C}{\sigma^2 n}} + 2\tilde{K}M\sqrt{\frac{\log C}{n}} + \sum_{k=1}^{K} R_k(\boldsymbol{\beta}^*)\,\mu_{kt}\right)$$

If $\sigma \sim \texttt{Inverse-Gamma}(a_0, b_0)$ and $\tau \sim \texttt{Half-Cauchy}(c_0, \gamma)$, then $E[1/\sigma] = a_0/b_0$ and $E[\sqrt{\tau}] = c_0 + \sqrt{2/\gamma}$. Since σ and τ are independent of each other, we can simplify $E(\tilde{Y}_t^{(0)} - \widehat{Y}_t)^2$ as,

$$E\left[E\left[\left(\tilde{Y}_t^{(0)} - \widehat{Y}_t\right)^2 \Big|\tau,\sigma\right]\right]$$
$$= \tilde{K}M\left(\frac{a_0}{b_0}(c_0 + \sqrt{2/\gamma})\sqrt{\frac{\rho(\boldsymbol{\beta}^*)\log C}{n}} + 2\tilde{K}M\sqrt{\frac{\log C}{n}} + \sum_{k=1}^{K} R_k(\boldsymbol{\beta}^*)\,\mu_{kt}\right)$$
$$\leq \tilde{K}\sqrt{\log n}\left(c_1\sqrt{\frac{\rho(\boldsymbol{\beta}^*)\log C}{n}} + 2\tilde{K}\sqrt{\log n}\sqrt{\frac{\log C}{n}} + \delta_n\|\boldsymbol{\mu}_k\|_1\right)$$
$$= O\left(\|\boldsymbol{\beta}^*\|_1\sqrt{\frac{\rho(\boldsymbol{\beta}^*)\log n\log C}{n}} + \|\boldsymbol{\beta}^*\|_1^2\sqrt{\frac{\log C(\log n)^2}{n}} + \sqrt{\log n}\|\boldsymbol{\beta}^*\|_1\delta_n\|\boldsymbol{\mu}_k\|_1\right)$$
$$= O\left(\|\boldsymbol{\beta}^*\|_1^2\sqrt{\frac{(\log C)^3}{n}} + \sqrt{\log n}\|\boldsymbol{\beta}^*\|_1\delta_n\|\boldsymbol{\mu}_k\|_1\right),$$

where the last inequalities follows since $C \geq n$, $\sqrt{\rho(\boldsymbol{\beta}^*)} \overset{O}{=} \|\boldsymbol{\beta}^*\|_1$ and $\log n$ dominates $\sqrt{\log n}$. Note that this simplification only gives us the error between $\widehat{Y}_t$ and $\tilde{Y}_t^{(0)}$, while we want to finally compute the error between $\widehat{Y}_t$ and $Y_t^{(0)}$. Thus, next we consider $E(Y_t^{(0)} - \tilde{Y}_t^{(0)})^2$ which constitutes the random error on the final prediction.

Note that based on our factor model, $Y_t^{(0)} - \tilde{Y}_t^{(0)} = \sum_{k=1}^{K} R_k(\boldsymbol{\beta}^*)\,\mu_{kt} + \rho(\boldsymbol{\beta}^*)\,\zeta_t \leq \delta_n\|\boldsymbol{\mu}_k\|_1 + \rho(\boldsymbol{\beta}^*)\,\zeta_t$. Thus, $E(Y_t^{(0)} - \tilde{Y}_t^{(0)})^2 \leq \delta_n^2\|\boldsymbol{\mu}_k\|_1^2 + \rho(\boldsymbol{\beta}^*)^2$. Combined with the simplification of $E(\tilde{Y}_t^{(0)} - \widehat{Y}_t)^2$, this proves our final bound since

$$E\left(Y_t^{(0)} - \widehat{Y}_t\right)^2 \leq E\left(\tilde{Y}_t^{(0)} - \widehat{Y}_t\right)^2 + E\left(Y_t^{(0)} - \tilde{Y}_t^{(0)}\right)^2.$$

C.3. Proof of Theorem 3

We first describe some properties pertaining to our choice of prior in our model coefficients β_c. The coefficients are mean zero Gaussian distributed with the variances being controlled by τ and λ_c^2. While τ is distributed with a Half Cauchy prior, λ_c's are Logit Normals with hyperpaparmeter as defined Lemma 4.

Lemma 5 *Consider the full prior on β_c described with $g(\beta_c)$ where, $\left[\beta_c|\tau,\lambda_c\right] \overset{i.i.d.}{\sim} N(0,\tau\lambda_c^2)$, $\lambda_c \overset{i.i.d.}{\sim}$ $\texttt{LogitNormal}(-Cf(C), C^2)$, $\tau \sim \texttt{HalfCauchy}(c_0,\gamma)$. Then, for $a_n = \sqrt{\frac{4\gamma(1+u)\log C}{nC^2}}$,*

$$1 - \int_{-a_n}^{a_n} g(x)dx \leq C^{-(1+u)} \;\; \textit{and} \;\; -\log\left(\inf_{x\in(-n,n)} g(x)\right) = O(\log C)$$

In addition to the properties on $g(\beta_c)$, in our case, we also have,

1.The scaled covariates $X_{ct}/\sqrt{\log nC}$ are bounded by M.
2.The dimensionality is high, $C \geq n$ and $l_n \log C \leq n$.
3.$\max\{|\beta_c^*/\sigma^*|\} \leq c_3 n$ for some fixed c_3.

The first statement holds since X_{ct}'s have nC Gaussian elements whose maximum is of the order $\sqrt{\log nC}$ and the other two statements follow from Assumption 1. Lemma 5 together with the above three assumptions are similar to the conditions required for Theorem 2.2 in Song and Liang (2023). The only difference being the first assumption, where Song and Liang (2023) assumes bounded covariates rather than the bound on scaled covariates.

Define $A_n = \{$ At least $\tilde{C}$ entries of $|\boldsymbol{\beta}/\sigma|$ is larger than $a_n\} \cup \{\left(\left\|\boldsymbol{x}'_{st}\boldsymbol{\beta}_s - \boldsymbol{x}'_{st}\boldsymbol{b}^*_s\right\| \geq c_1\sigma^*\epsilon_n\right\} \cup \{\sigma^2/\sigma^{*2} > (1+\epsilon_n)/(1-\epsilon_n)$ or $\sigma^2/\sigma^{*2} < (1-\epsilon_n)/(1+\epsilon_n)\}$, $B_n = \{$ At least $\tilde{C}$ entries of $|\boldsymbol{\beta}/\sigma|$ is larger than $a_n\}$ and $C_n = A_n \backslash B_n$, where $\tilde{C}$ is defined exactly as in Song and Liang (2023), $\tilde{C} = \lfloor \min\{\hat{c}_3, \tilde{c}_3\} n\epsilon_n^2/(2\log C)\rfloor$.

Following the three-step technique as in Theorem 2.2 in Song and Liang (2023) with the following testing functions completes the proof. Since the proof is similar, the details are omitted. Consider the following two testing functions,

$$\phi'_n = \max_{\{\xi \supseteq \xi^*, |\xi| \leq \tilde{C}+l_n\}} 1\{\left|\boldsymbol{y}^T(I-H_\xi)\boldsymbol{y}/(n-|\xi|)\sigma^{*2} - 1\right| \geq \epsilon_n\}, \text{ and}$$
$$\tilde{\phi}_n = \max_{\{\xi \supseteq \xi^*, |\xi| \leq \tilde{C}+l_n\}} 1\{\|X_\xi(X_\xi^T X_\xi)^{-1}X_\xi^T y - X_\xi\boldsymbol{\beta}^*_\xi)\| \geq c_1\sigma^*\epsilon_n\},$$

where $H_\xi = X_\xi\left(X_\xi^T X_\xi\right)^{-1} X_\xi^T$.

With these testing functions, we finally get the first main result since $y := \boldsymbol{x}'_{st}\boldsymbol{\beta}_s$,

$$P\left(\{\pi\left(\left\|\boldsymbol{x}'_{st}\boldsymbol{\beta}_s - \boldsymbol{x}'_{st}\boldsymbol{\beta}^*_s\right\| \geq c_1\epsilon_n \middle| \mathcal{D}_n(s)\right)\} \geq e^{-c_2\, n\, \epsilon_n^2}\right) \leq \exp(-c_3\, n\, \epsilon_n^2)\,. \tag{6}$$

Now based on (2) in Lemma 1, we know that for any time $t > n$,

$$Y_{st}^{(0)} = \boldsymbol{x}'_{st}\boldsymbol{\beta}^*_s + \sum_{k=1}^{K} R_{sk}(\boldsymbol{\beta}^*_s)\mu_{kst} + \rho(\boldsymbol{\beta}^*_s)\zeta_{st}\,.$$

With the decomposition of the counterfactual, it is easy to note that if $\|\boldsymbol{x}'_{st}\boldsymbol{\beta}_s - \boldsymbol{x}'_{st}\boldsymbol{\beta}^*_s\| \leq c_1\epsilon_n$, then

$$\begin{aligned}\left\|y - Y_{st}^{(0)}\right\| &\leq \left\|y - \boldsymbol{x}'_{st}\boldsymbol{\beta}^*_s\right\| + \left\|\sum_{k=1}^{K} R_{sk}(\boldsymbol{\beta}^*_s)\mu_{kst}\right\| + \left\|\rho(\boldsymbol{\beta}^*_s)\zeta_{st}\right\| \\ &\leq c_1\epsilon_n + \delta_n\|\boldsymbol{\mu}_{st}\|_1 + \rho(\boldsymbol{\beta}^*_s)|\zeta_{st}|\,.\end{aligned} \tag{7}$$

Consider the set, $\mathcal{S} := \{s : \pi\left(\left\|y - \boldsymbol{x}'_{st}\boldsymbol{\beta}^*_s\right\| \geq c_1\epsilon_n \middle| \mathcal{D}_n(s)\right) \leq e^{-c_2\, n\, \epsilon_n^2}\}$, which gives us $P(\mathcal{S}) \geq 1 - \exp(-c_3 n\epsilon_n^2)$. Based on the definition, under the set $\mathcal{S}$, we have $\pi\left(\left\|y - \boldsymbol{x}'_{st}\boldsymbol{\beta}^*_s\right\| \geq c_1\epsilon_n\right) \leq e^{-c_2\, n\, \epsilon_n^2}$ which is equivalent to $\pi\left(\left\|y - \boldsymbol{x}'_{st}\boldsymbol{\beta}^*_s\right\| \leq c_1\epsilon_n\right) \geq 1 - e^{-c_2\, n\, \epsilon_n^2}$. As shown above in (7), since $\left\|y - \boldsymbol{x}'_{st}\boldsymbol{\beta}^*_s\right\| \leq c_1\epsilon_n$ implies $\left\|y - Y_{st}^{(0)}\right\| \leq c_1\epsilon_n + \delta_n\|\boldsymbol{\mu}_{st}\|_1 + \rho(\boldsymbol{\beta}^*_s)|\zeta_{st}|$, under $\mathcal{S}$, we have

$$\pi\left(\left\|y - Y_{st}^{(0)}\right\| \leq c_1\epsilon_n + \delta_n\|\boldsymbol{\mu}_{st}\|_1 + \rho(\boldsymbol{\beta}^*_s)|\zeta_{st}|\right) \geq 1 - e^{-c_2\, n\, \epsilon_n^2}\,.$$

Finally, since ζ_{st} is standard Gaussian, hence $P(|\zeta_{st}| \leq c_5) \geq 1 - 2\exp(-c_5^2/2)$. Combining this with the above bound on the posterior distribution finally gives us that under $\mathcal{S}$ and subsequently with probability $1 - \exp(-c_3 n\epsilon_n^2)$,

$$\pi\left(\left\|y - Y_{st}^{(0)}\right\| \geq c_1\epsilon_n + \delta_n\|\boldsymbol{\mu}_{st}\|_1 + \rho(\boldsymbol{\beta}^*_s)c_5 \middle| D_n(s)\right) \leq \exp(-c_2\, n\, \epsilon_n^2) + 2\exp(-c_5^2/2)\,,$$

The trimmed mean result follows directly by considering the $2(\exp(-c_2\, n\, \epsilon_n^2) + 2\exp(-c_5^2/2))$ trimming of the posterior distribution in the set $\mathcal{S}$.

C.4. Proof of Lemma 4

We have $X \sim \texttt{LogitNormal}(-tf(t), t^2)$, and $\exp(f(t)^2/2)f(t) = t^{(1+u)}$.

$$\begin{aligned} P[X \in (0, t^{-1/2})] &= P[\log X \in (-\infty, -0.5\log t)] \\ &\approx P[\text{logit}(X) \in (-\infty, -0.5\log t)] \\ &= P\left[\frac{\text{logit}(X) + tf(t)}{t} \in \left(-\infty, -\frac{\log t}{2t} + f(t)\right)\right] \\ &= \Phi\left(-\frac{\log t}{2t} + f(t)\right) \approx \Phi\left(f(t)\right) \end{aligned}$$

The second inequality holds, since for large t, in the set $\Theta = (0, t^{-1/2})$, $\theta \in \Theta$ $\text{logit}(\theta) = \log(\theta) - \log(1-\theta) \approx \log(\theta)$ as $\log(1-\theta)$ is close to 0 since θ as $t^{-1/2}$ is small. The last step follows as $\text{logit}(X) \sim \mathcal{N}(-tf(t), t^2)$ and consequently $t^{-1}(\text{logit}(X) + tf(t))$ is standard normal.

Using Mill's inequality, $1 - \Phi\left(f(t)\right) = O(\exp(-f(t)^2/2)/f(t)) = O(t^{-(1+u)})$ or $1 - \Phi\left(f(t)\right) = k_1 t^{-(1+u)}$ for some constant k_1 which proves the required.

The second one follows similarly as $P[X \in (1 - t^{-1/2}, 1)] = \Phi\left(\frac{\log t}{2t} + f(t)\right) \approx \Phi\left(f(t)\right)$.

C.5. Proof of Lemma 5

For any set of hyperparameters τ, λ_c, the β_c is zero mean Gaussian distribution. Thus $\inf_{x\in(-n,n)} g(x)$ is $g(n)$ since the marginal distribution is symmetric around zero and keeps decreasing as x increases.

Consider the set $\mathcal{A} = \{\tau \geq \gamma C^2 + c_0\} \cap \{\lambda_c \geq 1 - C^{-1/2}\}$. In this set A, variance $\tau\lambda_c^2 \geq (\gamma C^2 + c_0)(1 - C^{-1/2})^2 \geq \gamma C^2/2$ for $C > 1$. Under this set $\mathcal{A}$, $g(n) \geq (C\sqrt{\pi})^{-1} C \exp(-n^2/\gamma C^2)$. Also the probability $P(\mathcal{A}) = P(\tau \geq \gamma C^2 + c_0)P(\lambda_c \geq 1 - C^{-1/2}) \approx (2/\pi C^2)C^{-(1+u)} = 2C^{-(3+u)}/\pi$ since τ and λ_c are independent.

Using the fact that $P(\mathcal{A}) \approx 2C^{-(3+u)}$ and $g(n|\mathcal{A}) \geq (C\sqrt{\pi})^{-1}\exp(-n^2/\gamma C^2)$, thus the marginal distribution g is such that

$$\begin{aligned} g(n) \geq P(\mathcal{A}g(n|\mathcal{A})) &\approx \frac{2}{\sqrt{\pi}} C^{-(4+u)} \exp(-n^2/\gamma C^2) \\ \implies -\log(g(n)) &\leq \frac{n^2}{\gamma C^2} + (4+u)\log C - \log(2/\sqrt{\pi}) = O(\log C)\,, \end{aligned}$$

which follows since $C \geq n$.

For the second inequality consider the set $\mathcal{B} = \{\tau \leq c_0 + \gamma C^{1+u}\} \cap \{\lambda_c \leq C^{-(4+u)/2}\}$. The probability of this set, $P(\mathcal{B}) = P(\tau \leq c_0 + C^{1+u})P(\lambda_c \leq C^{-(4+u)/2})$. Consider $P(\lambda_c \leq C^{-(4+u)/2})$,

$$\begin{aligned} P[\lambda_c \leq C^{-(4+u)/2})] &= P[\log \lambda_c \leq -(4+u)\log C/2] \approx P[\text{logit}(\lambda_c) \leq -(4+u)\log C/2] \\ &= P\left[\frac{\text{logit}(\lambda_c) + Cf(C)}{C} \leq -\frac{(4+u)\log C}{2C} + f(C)\right] \\ &= \Phi\left(-\frac{(4+u)\log C}{2C} + f(C)\right) \approx \Phi\left(f(C)\right) = 1 - c_0 C^{-(1+u)}\,, \end{aligned}$$

for some constant c_0 and, where the last inequality follows from Mill's inequality and $\exp(f(C)^2/2)f(C) = t^{(1+u)}$. Also, $P(\tau \leq c_0 + C^{1+u}) \approx 1 - 2C^{-(1+u)}/\pi$. Thus,

$$\begin{aligned} P(\mathcal{B}) &= P(\tau \leq c_0 + C^{1+u})P(\lambda_c \leq C^{-(4+u)/2}) \\ &= (1 - c_0 C^{-(1+u)})(1 - 2C^{-(1+u)}/\pi) \approx 1 - (2/\pi + c_0)C^{-(1+u)}\,. \end{aligned}$$

Also, in this set $\mathcal{B}$, the marginal variance of β_c is $\tau\lambda_c^2 \leq (c_0 + \gamma C^{1+u})C^{-(4+u)} \leq 2\gamma C^{-3}$. Thus, if $(\tau, \lambda_c) \in \mathcal{B}$, then $\int_{a_n}^{\infty} g(x)dx \leq \int_{a_n}^{\infty} f(x)dx$, where f is density of Gaussian distribution with variance $2\gamma C^{-3}$.

$$\int_{a_n}^{\infty} g(x)dx \leq \int_{a_n}^{\infty} f(x)dx = O(\exp(-a_n^2/2\sigma^2)/(a_n/\sigma))$$

where $\sigma = \tau\lambda_c^2$. Since $a_n^2/\tau\lambda_c^2 \geq (4\gamma(1+u)\log C/nC^2)(2\gamma C^{-3})^{-1} \geq 2(1+u)\log C$, as $C \geq n$. Thus $O(\exp(-a_n^2/2\sigma^2)/(a_n/\sigma)) = O(C^{-(1+u)}/\sqrt{\log C}) \leq C^{-(1+u)}$ for large enough C, which completes the proof.

C.6. Summary of notation used in Section 3

Table 7. Summary of notation used in the methodology section.

Notation	*Description*
c, C	*c indexes control drugs, $c = 1, \ldots, C$, where C is the total number of control drugs*
s, S	*s indexes states, $s = 1, \ldots, S$, where S is the number of states*
t, N	*t indexes for time (weeks), $t = 1, \ldots, N$, where N is the total number of weeks*
X_{cst}, $\boldsymbol{x}_{st}$	*X_{cst} denotes prescription claims of control drug c in state s at time t, $\boldsymbol{x}_{st} = (X_{cst} : 1 \leq c \leq C)$*
n	*Number of pre-treatment weeks*
Y_{st}	*Observed IVM prescription claims in state s at time t in the pre-treatment period ($t \leq n$)*
$Y_{st}^{(1)}$	*Observed IVM claims in state s at time t in the post-treatment period ($t > n$)*
$Y_{st}^{(0)}$	*Counterfactual IVM claims without intervention in state s at time t, $t > n$*
$\widehat{Y}_{st}^{(0)}$	*Estimated counterfactual IVM claims in state s at time t, $t > n$*
$\widehat{ATT}_{st}$	*Estimated average treatment effect on the treated $Y_{st}^{(1)} - \widehat{Y}_{st}^{(0)}$*
k, K	*k indexes latent factors, $k = 1, \ldots, K$, where K is the total number of latent factors*
ϕ_{sk}	*Factor loading for state s and factor k*
μ_{kst}, $\boldsymbol{\mu}_{st}$	*μ_{kst} denotes latent factor value for factor k in state s at time t, $\boldsymbol{\mu}_{st} = (\mu_{kst} : 1 \leq k \leq K)$*
ε_{st}	*Noise term in treated outcome model for Y_{st}, $E(\varepsilon_{st}) = 0$*
ψ_{csk}, $\boldsymbol{\psi}_{cs}$	*ψ_{csk} denotes factor loading for control unit c in state s at time t, $\boldsymbol{\psi}_{cs} = (\psi_{csk} : 1 \leq k \leq K)$*
ε_{cst}	*Noise term in control outcome model for X_{cst}, $E(\varepsilon_{cst}) = 0$*
β_{cs}, $\boldsymbol{\beta}_s$	*β_{cs} synthetic control weights for state s on control c, $\boldsymbol{\beta}_s = (\beta_{cs} : 1 \leq c \leq C)$*
$\widehat{Y}_{st}(\boldsymbol{\beta}_s)$	*Linear synthetic control prediction based on $\boldsymbol{\beta}_s$, $\boldsymbol{\beta}_s'\boldsymbol{x}_{st}$*
$R_{sk}(\boldsymbol{b})$, $R_s(\boldsymbol{b})$	*$R_{sk}(\boldsymbol{b})$ denotes residual factor loading difference for any $\boldsymbol{b} \in \mathbb{R}^C$, $\phi_{sk} - \sum_{c=1}^{C} b_c\psi_{csk}$ and $R_s(\boldsymbol{b}) = (R_{sk}(\boldsymbol{b}) : q \leq k \leq K)$*
τ	*Global shrinkage parameter in Logit-Normal prior (LNP)*
λ_{cs}	*Local shrinkage parameter in Logit-Normal prior (LNP) for control c and state s*
m, v	*Hyper-parameters for the local shrinkage parameter λ_{cs} in Logit-Normal prior (LNP)*
$\boldsymbol{\beta}_s^*$,	*Sparse true coefficient vector $\in \mathbb{R}^C$*
δ_n	*Maximum residual factor loading error*
$\widehat{\boldsymbol{\beta}}_s[m]$, $\widehat{\boldsymbol{\beta}}_s[mo]$	*$\widehat{\boldsymbol{\beta}}_s[m]$ denotes the posterior mean of $\boldsymbol{\beta}_s$, $\widehat{\boldsymbol{\beta}}_s[mo]$ denotes the posterior mode of $\boldsymbol{\beta}_s$*
$\widehat{Y}_{st}^{(0)}[m]$,$\widehat{Y}_{st}^{(0)}[mo]$	*Synthetic control prediction based on $\widehat{\boldsymbol{\beta}}_s[m]$, $\widehat{\boldsymbol{\beta}}_s[mo]$ respectively*
$\mathcal{R}_n^{(i)}$	*Components of prediction error bound $i = 1, \ldots, 4$*

D. Further discussion of state-level heterogeneity in IVM consumption

To further illustrate the heterogeneity in IVM consumption across states, we examine two large states with contrasting political leanings: Texas, a Red state, and California, a Blue state (Figure 12). The time series of IVM prescription claims shows broadly similar temporal patterns across these politically diverse states, suggesting common underlying drivers of demand. However, Texas consistently exhibits higher per capita IVM prescription claims than California throughout the observation period, despite having lower COVID-19 cases per 100,000 people. This disparity between prescription intensity and disease prevalence suggests that factors beyond epidemiological conditions may have influenced IVM consumption.

Figure 13 further illustrates these differences by plotting ATT per capita for California and Texas as representative Blue and Red states. Prior to the period when IVM gained significant public attention, the ATT per 100,000 people remains close to zero in both states. Following this period, however, the increase in prescription claims becomes systematically larger in Texas than in California. While the two states display broadly similar temporal patterns, the magnitude of the increase differs substantially. In particular, after Dr. Kory's testimony on December 8, 2020, Texas shows a markedly larger rise in IVM prescription claims than California, despite having lower COVID-19 case rates per capita (see Figure 12, bottom panel).

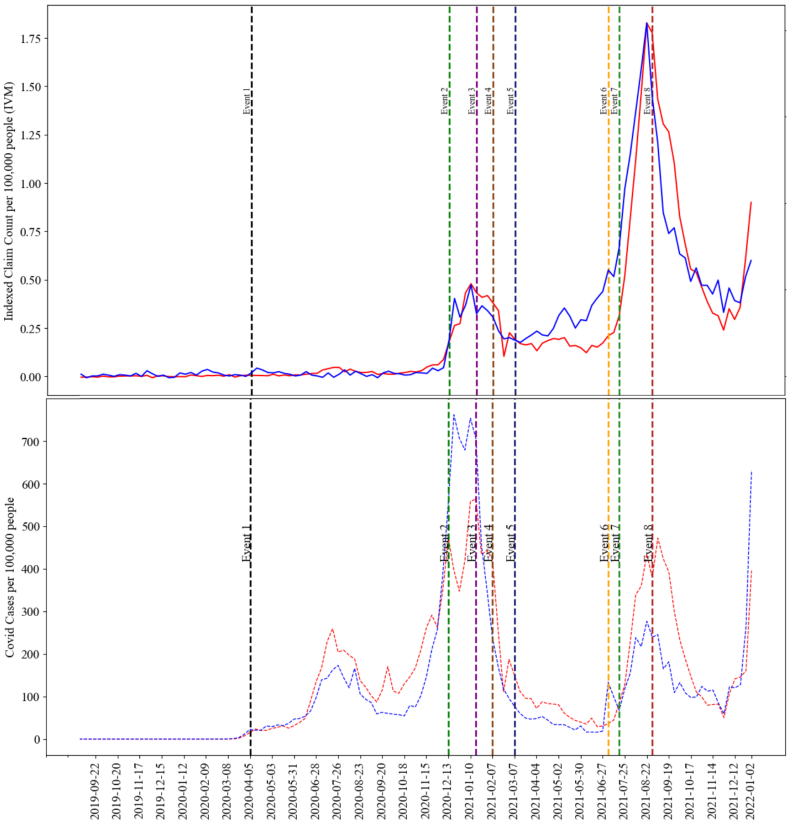

Fig. 12: Plot of weekly per capita IVM prescription claims (top) and Covid case counts (bottom) in California (in blue) and Texas (in red).

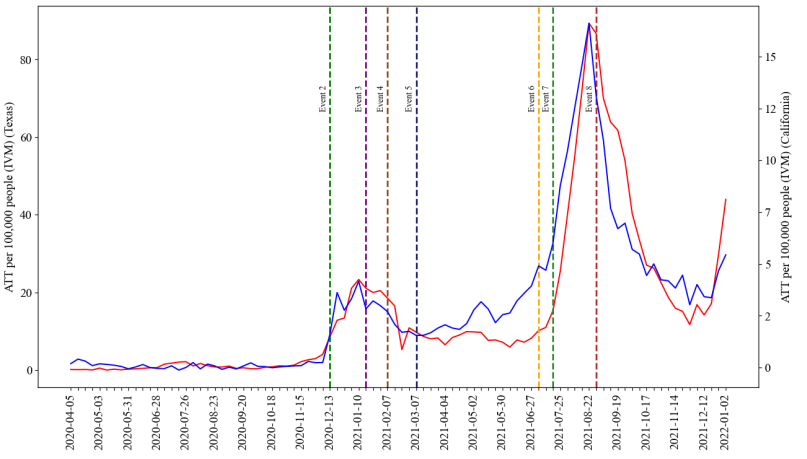

Fig. 13: Plot of weekly ATT estimates for IVM prescription claims for California (in blue) and Texas (in red). The time points marked by dotted vertical lines correspond to major related events listed in Table 1.

8. Competing interests

No competing interest is declared.

9. Author contributions statement

D.P. and I.B. conducted the data analysis. R.R.B. and G.M. developed the statistical methodology and provided theoretical proofs related to the synthetic control approach. S.D., J.M., and R.M. were responsible for data extraction and preprocessing. D.P., R.R.B., S.D., and G.M. contributed to the writing of the manuscript.

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